Taiwan Semiconductor Manufacturing Company

Full investment research | Model assumptions and source evidence

1amInvesting | Full company research | Framework 8.0

Taiwan Semiconductor Manufacturing Company

TSM | NYSE ADS; underlying 2330 Taiwan Stock Exchange | 2026-10-15

Q4 2027 model ceiling: USD 624.40

Analytical Base 50/60 = 83 | High Conviction / structural qualified-capacity bottleneck

20x forward-four-quarter normalized IFRS operating profit attributable to parent, plus fixed opening surplus liquidity less face debt, lease and pension claims, divided by end-window ADS-equivalent shares. Each ADS represents five ordinary shares; Q prices Q plus the next three quarters.

Coverage initiated at the owner's request. No holding was established by this research. The portfolio Base remains 83; no manual override is introduced.

All operating values are in USD million and shares in millions unless stated otherwise. Forecasts are analytical assumptions; issuer guidance and actual results are identified separately.

Writing-date snapshot: USD 472.20 per share (2026-10-07 close); 5,186.474 million full-company ADS equivalents (June30 full-company ADS-equivalent); estimated equity capitalization USD 2,449,053.0 million. ADR-equivalent capitalization; not the number of traded depositary receipts. Vendors differ by a few cents at the close; use the $472.20 official-close series consistently. This dated ADS-equivalent snapshot is separate from forecast diluted shares.

October 15 earnings and guidance: current decision

TSMC has published genuine Q3 results and Q4 guidance. Q3 revenue is **USD 46.72 billion**, gross profit **USD 31.028 billion**, R&D **USD 2.782 billion** and SG&A **USD 0.859 billion**. Reported operating profit of USD 27.403 billion includes USD 0.016 billion other operating income. The retained normalized convention excludes that separate item: GP minus R&D and SG&A is USD 27.387 billion, then the unchanged 0.1% minority operating-profit reserve gives **USD 27.359613 billion** attributable to common owners. The issuer's USD 9 million noncontrolling net income is after tax and does not establish an operating-profit allocation. Q3 results

Q4 uses the midpoint of issuer revenue guidance of USD 52.3-54.3 billion: **USD 53.3 billion**, with 66% gross margin and 57.5% normalized consolidated operating margin. These are midpoint model selections within guidance, not promised results. Costs total 8.5% of sales; their R&D/SG&A split retains the prior modeled proportions. The issuer's guidance FX is 31.8; the USD operating anchors are used directly while the separate valuation translation remains 32.

The active **Q4 2027 ceiling remains USD 624.40 per ADS**, using Q4 2027 through Q3 2028 earnings. These new Q3/Q4 2026 inputs lie outside that window. The 2027+ operating rows, five forward probabilities, 32 joint states, fixed opening financial-asset credit, end-window shares and both Bases83 are unchanged. Nearer Q3/Q4 2026 ceilings become USD 459.22/USD 485.94. This is an earnings update with an unchanged active-horizon target.

The refreshed 2026 revenue bridge is USD 176.12 billion using the retained rounded H1 issuer anchors and Q4 midpoint; the existing 2027 model already grows 28.5% from that bridge. Strong demand and steep N2 ramp are consistent with an existing large growth assumption. No new paid wafer capacity, eligible contract pricing, customer invoice capture or outyear cost bridge is disclosed in the retrieved release/deck/report that supports an incremental later-year uplift. Existing nonzero

pricing, CPU and COUPE branches remain active; this avoids counting their economics twice.

Dated history, October 11: July-September monthly disclosures had already supplied NT$ 1,494.243284 billion, translated at model FX32 into USD 46.695103 billion. The new quarter's NT$ 1,494.243 billion differs only by presentation rounding; the USD 24.897375 million revenue difference is principally translation. The whole monthly-sales beat is not applied a second time.

Decision in the model

Active horizon	Earnings window	F4Q OP	Multiple	Diluted shares	Ceiling
Q4 2027	Q4 2027 through Q3 2028	160,972.1	20x	5,198.151	624.40

The ceiling is a model valuation, not a guaranteed trading price or a literal net-income P/E. Operating branches are weighted once. Where debt, preferred claims, conversion or equity floors make value nonlinear, the model values each coherent operating and funding state first, then weights those common-share outcomes.

Economic scope and matching claims

Consolidated wafer and backend service revenue; ordinary depreciation, amortization and awards remain expenses. Nonoperating VIS gains excluded. A rising minority earnings reserve is applied before valuation. 20x forward-four-quarter normalized IFRS operating profit attributable to parent, plus fixed opening surplus liquidity less face debt, lease and pension claims, divided by end-window ADS-equivalent shares. Each ADS represents five ordinary shares; Q prices Q plus the next three quarters.

Financial claims and diluted shares follow the company-specific schedule below. Debt, preferred and minority claims are counted once. All financial model tables use USD million unless a different unit is stated.

Investment conclusion

TSMC combines scarce manufacturing processes with the design libraries, production experience and packaging needed to turn a chip design into reliable shipped products. Its customer base spans competing GPU, custom-accelerator and CPU architectures. That broad exposure makes it an attractive way to own growing silicon complexity, while avoiding a forecast that requires one architecture to win every workload.

The active model produces a Q4 2027 ceiling of **USD 624.40 per ADS** at **20x forward-four-quarter normalized operating profit**, with an analytical **Base of 83**. The earnings window is Q4 2027 through Q3 2028. This is the framework's operating-profit convention, not a 20x net-income P/E. The largest positive drivers are leading-node volume, backend services and moderate repricing. N2 qualification, overseas costs, customer deployment timing and second sourcing remain explicit risks.

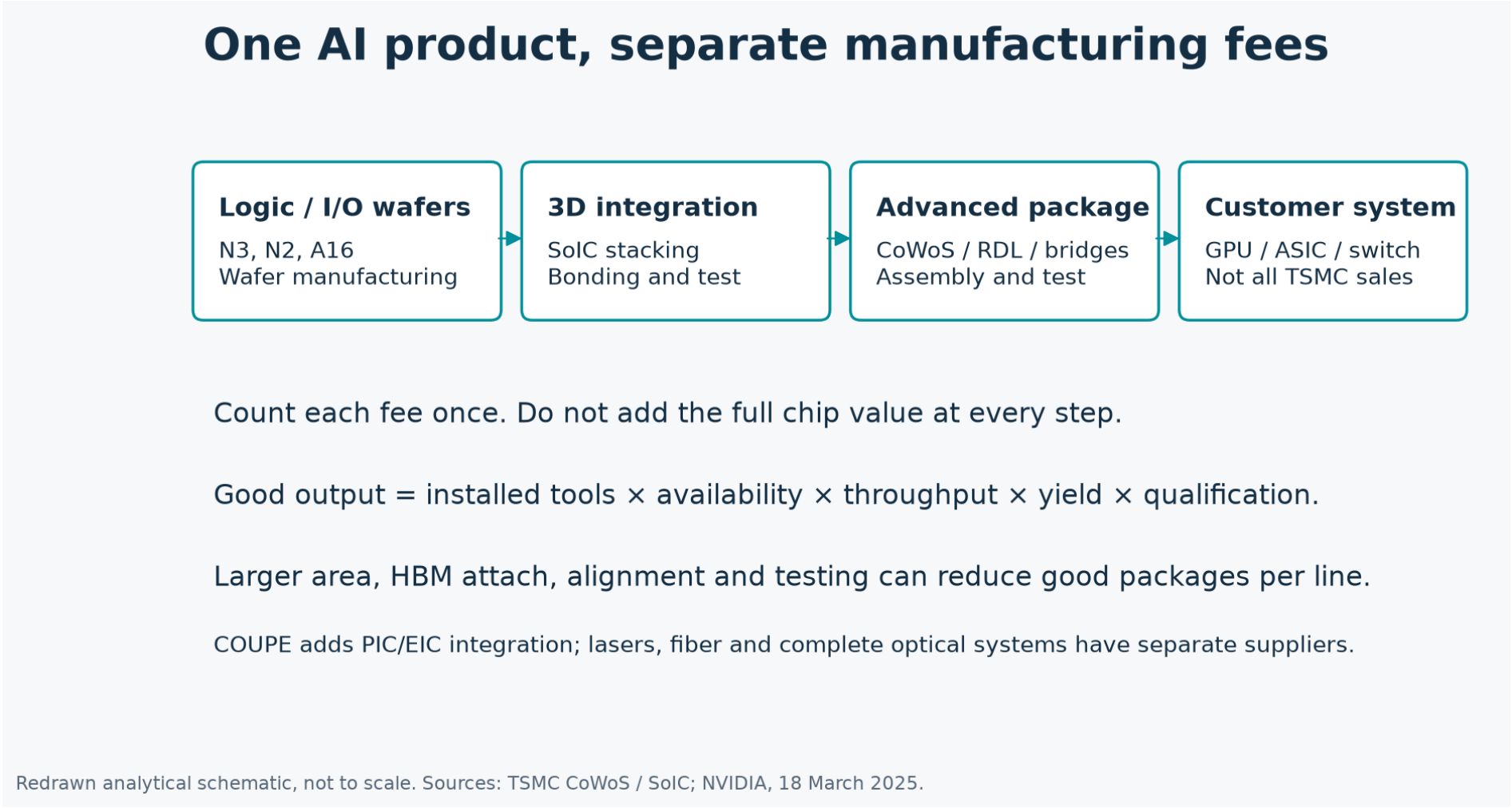
This is a strong business with an already visible moat. The forecast does not require monopoly pricing, uninterrupted AI spending or the elimination of competing foundries. Its more demanding assumption is that qualified output and customer deployments can support annual sales around USD 226.3 billion in 2027 and USD 290.7 billion in 2028. Beyond 2028, the forecast is a capacity and economics scenario with lower visibility.

What the company sells

TSMC manufactures customer-designed chips and provides related masks, packaging and testing. A customer pays for its manufacturing and service content; TSMC does not collect the whole selling price of a GPU, server or cloud service. A system with more HBM, optical components or cooling creates TSMC revenue only where TSMC supplies the associated logic, bonding or package services.

The revenue model separates wafer fabrication from non-wafer services. Within wafers it distinguishes N2/A16, N3, N5/N7, mature/specialty and later A14-family products. Platform categories such as high-performance computing and smartphones provide a customer-demand cross-check. They are another view of the same company sales, not additional revenue pools.

Q2 accounts report NT$ 1,073.7 billion of wafer revenue and NT$ 196.7 billion of other revenue. The latter is broader than advanced packaging: it includes testing, masks and services. The model therefore uses a16% other-revenue share inQ3, rising to 18% in 2027 and 20% in 2028. Within that total, advanced packaging/testing is a 13%/15%/17% company-sales proxy for 2026-28; roughly 3% is other non-wafer work. Q3 actual wafer technology shares are 8% N2,33% N3,38% N5/N7 and 21% mature/specialty. They are applied inside the retained 16% non-wafer allocation, which remains a model proxy because the Q3 summaries do not disclose the actual wafer/non-wafer dollar split. Node gross margins are uniformly recalibrated to the reported group gross profit; individual node margins are not issuer disclosures. Interim accounts



Analytical schematic | Wafer, stacking and packaging fees are separate; customer product value is counted once.

The manufacturing moat

A process is commercially valuable when a customer can design with it, achieve acceptable yield and reliability, and obtain enough qualified production at a useful cost. A transistor demonstration is only part of that chain. Process design kits, libraries, physical implementation, masks, wafer processing, package design and test must work together. Requalifying a complex chip at another foundry takes engineering effort and time, and can require a redesign.

TSMC's scale spreads development expense across many products and customers. Its independence as a manufacturing partner also allows competing designers to share a production ecosystem. These advantages support sustained returns, but they do not make all proposed capacity profitable. A poor customer design can have

weak yield even on a capable process; conversely, an excellent test structure does not prove that a very large customer's die has reached its target yield.

For this reason, the Jalapeño yield discussion is a program risk, not measured evidence of a systemic TSMC process failure. The archived engineering critiques do not provide a sampled distribution of wafers, bins, die sizes and process versions. The model retains joint execution risk rather than assigning the whole foundry a new unmeasured yield penalty.

Process and packaging roadmap

The April symposium places A14 volume production in 2028, N2U in 2028, and A13/A12 in 2029. A13 is an optical shrink of A14; A12 adds backside power delivery. These are roadmap commitments, not current recognized sales. Device performance, power and density comparisons have different conditions and cannot be added together as a single performance gain. TSMC symposium

Packaging makes a large chip system manufacturable by joining compute dies and memory with dense connections. CoWoS, SoIC and optical integration address different parts of that system. The symposium reports5.5-reticle CoWoS in production,14-reticle packages planned for 2028, and larger systems later. A bigger package can contain more compute and memory while consuming more substrate area, bonding operations and test time. Nominal wafer capacity therefore does not translate one-for-one into finished systems.

The Q1 call described N2's initial production ramp and planned N3 additions in Taiwan, Arizona and Japan. Converting N5 equipment to N3 reallocates tools; it is not simultaneous new capacity in both nodes. Capital spending must precede installation, qualification, utilization and accepted shipments. Q1 management call

TSMC: production status versus roadmap

2025 reported	**N2 volume production begins in Q4** No disclosed large AI-die yield percentage
2026 status / plan	**N2P / A16 H2; 5.5-reticle CoWoS; COUPE** Separate product families and capacity constraints
2027 attributed	**COUPE 3.2 → 6.4 Tbps per engine** TrendForce public preview; no disclosed unit / fee contract
2028 roadmap	**A14 / N2U; 14-reticle CoWoS** Customer qualification and good-unit output still required
2029 roadmap	**A13 / A12; 40-reticle SoW-X** Integration scale is not a same-priced chip sales multiplier

Redrawn: issuer technology pages / April 2026 symposium; TrendForce public preview, 7 October 2026. Plans can change.

Analytical source summary | Reported production and future roadmap status are distinguished.

Actual earnings and the forecast starting point

FY2025 revenue was USD 122.42 billion. Q1/Q2 2026 revenue was USD 35.90/40.20 billion. Q3 has now been reported at USD 46.72 billion, NT$ 1,494.243 billion. The financial statements use an average31.983NTD/USD for Q3; the balance sheet's convenience translation uses31.880 and cash-flow USD columns describe nine-month totals at 31.744. These are different periods and scopes. Q3 financial statements

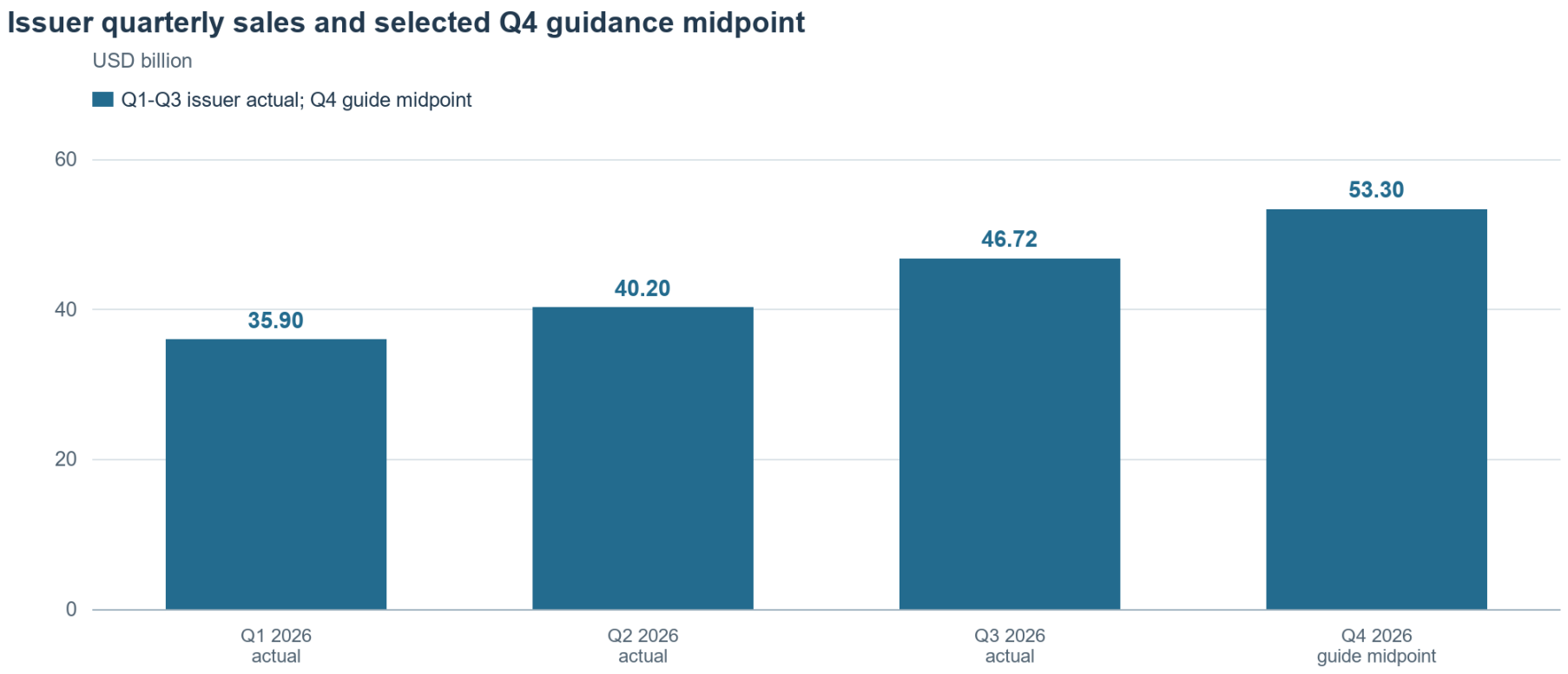
Q3 gross profit is USD 31.028 billion, R&D2.782 billion and SG&A0.859 billion. The group reported USD 27.403 billion OP, including USD 0.016 billion other operating income; the model normalizes to USD 27.387 billion before the unchanged 0.1% minority proxy. Ordinary D&A and awards remain expenses. Nonoperating VIS investment gains are excluded. Actual net income attributable to parent is USD 23.858 billion and actual NCI net income USD 0.009 billion; neither is substituted for normalized OP.

Gross margin 66.4% fell 1.3 percentage points sequentially, primarily from the initial N2 ramp. Opex increased NT$ 17.45 billion. The absence of Q2 earthquake insurance reduced other operating income, and nonoperating gains fell as Q2 VIS disposal/mark-to-market gains did not recur. These are counterweights to the sales beat, not hidden recurring profit additions. Management report

Actual wafer revenue mix is N2/N3/N5/N7=8/33/28/10%; advanced technologies represent79% of wafer revenue. HPC/smartphone/IoT/auto/DCE/others are 68/21/4/4/1/2% of consolidated revenue. The two tables have different denominators and are not added together. Actual shipments are 4.476 million12-inch-equivalent wafers; that total does not establish qualified N3 capacity or a specific node's realized ASP.

Q4 revenue guidance is USD 52.3-54.3 billion, GM65-67% and OP margin 56.5-58.5% at 31.8NTD/USD. Midpoint selections yield sales 53.3 billion, GP35.178 billion, normalized consolidated OP30.6475 billion and total operating cost4.5305 billion. Parent OP is USD 30.6168525 billion after the existing0.1% reserve. Q4 node allocations and gross margins remain modeled and calibrated to group guidance, with no issuer-specific cost split asserted.

The monthly-sales history remains evidence of the same quarter: July/August/September NT$ 467,580,548/514,805,337/511,857,399 thousand sum to NT$ 1,494,243,284 thousand. Division by 1,000 then32 produced the earlier USD 46,695.102625 million estimate. It is historical reconciliation, not an additional economic event.



Q1-Q3 issuer USD revenue; Q4 is selected guidance midpoint. Q3 group GP/costs are actual; normalized OP excludes other operating income and uses the modeled minority reserve.

Customer demand and the revenue bridge

The reference trajectory is USD 227/288/340/380 billion of sales in 2027-30. It is a company manufacturing forecast, not a fixed percentage of global AI capital expenditure. The operating model assigns that wallet to distinct manufacturing vectors, then tests the implied output and mix. Advanced packaging/test reference revenue is an allocation proxy within the updated 2026 bridge, then rises to USD 34.1 billion in 2027 and USD 49.0 billion in 2028. Other non-wafer revenue is separate within the same consolidated total.

Management's latest discussion identifies agentic workloads as an additional CPU opportunity across architectures. The active case assigns60% probability to a bounded incremental foundry wallet: gross customer manufacturing demand ofUSD 8/14/18/20 billion in 2027-30,80% capture, and a50% deduction for demand already in the HPC reference. That leaves conditional annual sales ofUSD 3.2/5.6/7.2/8.0 billion before probability weighting. The allocation is an assumption, not management guidance.

TrendForce's October14 public research raises top-nine-CSP2027 capex growth from 50% to 60%, citing server/component price increases and supplier prepayments. It projects AI-server unit growth near36% versus production value rising from about USD 479 billion to 920 billion, or92.1%. The implied value-per-system increase is roughly41.2% if the scopes align, not a measured TSMC wafer-price increase. Rack mix, memory and prepayment timing matter. These forecasts corroborate demand/cost context; they do not establish additional TSMC retained invoices beyond existing capture and pricing assumptions. TrendForce October14

This approach avoids treating the migration from a merchant GPU to an in-house ASIC as an entirely new wafer sale. The same customer compute budget can fund either design. A change in die size, process node or packaging can increase TSMC content, but that increase must be demonstrated in the manufacturing invoice rather than inferred from the accelerator's retail price.

The following platform allocation is an independent check on the same reference revenue. It is not added to the node and backend model. The full-year2026 platform split remains an illustrative64%HPC/23%smartphone/13%other cross-check. It is not the actual quarter mix; Q3 reported 68%HPC and 21%smartphone. The 2027 HPC requirement grows about 41%, slowing thereafter; smartphones and other markets supply the rest. These company invoice pools avoid converting cloud capex into foundry revenue at an arbitrary constant ratio.

Year	Reference sales USD bn	HPC share assumption	HPC invoices USD bn	Smartphone invoices USD bn	Other invoices USD bn
2026	176.1	64%	112.7	40.5	22.9
2027	227.0	70%	158.9	43.1	25.0
2028	288.0	73%	210.2	49.0	28.8
2029	340.0	75%	255.0	54.4	30.6
2030	380.0	76%	288.8	57.0	34.2

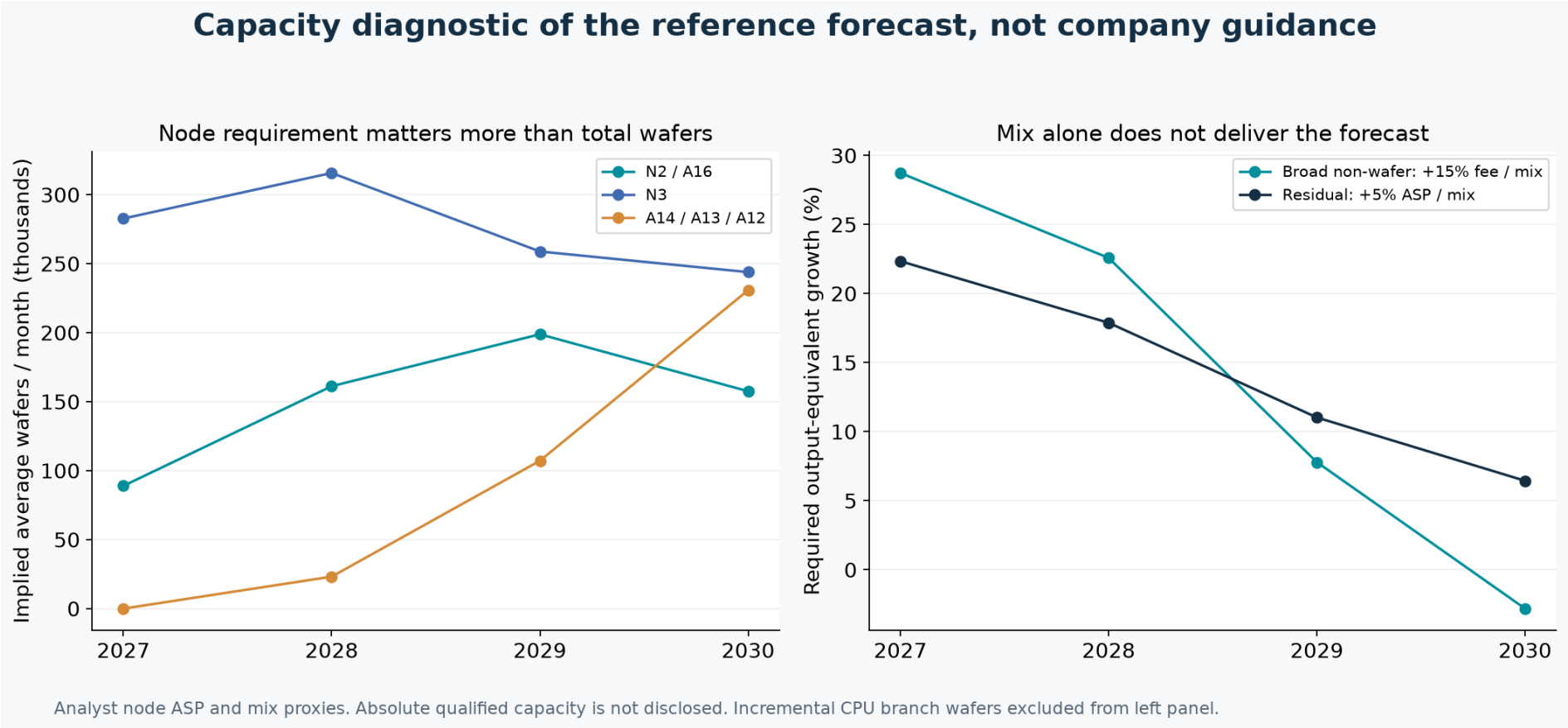
Capacity and the price of qualified output

Revenue can increase through more wafers, a richer node mix, more backend content and higher like-for-like prices. The model keeps those drivers separate. Its wafer ASP proxies are USD 30,000 for N2/A16,USD 22,000 for N3,USD 12,000 for the blended N5/N7 pool,USD 3,980 for mature/specialty equivalents andUSD 36,000 for the later A14 family. These are analytical conversion assumptions, not verified contract quotations. They are calibrated against reported equivalent shipments and must be revised if the scope proves inconsistent.

The N2/A16 reference share of wafer revenue rises through 14%,16%,18% and 20% in 2027, followed by 22%,24%,26% and 28% in 2028. The corresponding volume requirement is shown in the capacity exhibits. This slower early ramp makes N3 do more of the near-term work. It remains demanding: new N3 fabs, tool conversions and yield improvements must generate qualified output on time. A20% lower assumed wafer ASP means25% more wafers are needed for the same revenue; a20% higher ASP means about 17% fewer. This sensitivity is not proof that capacity exists.

October 6 channel relays suggested5-10% advanced-node price increases, but their numerical original images were unavailable. They are useful forward evidence with a wider uncertainty band. The active price branch uses 65% probability and only 50% eligible/repriceable invoice coverage at a6% conditional increase in 2027, equivalent to 3% on reference N2/N3 sales. A later repost is not a second pricing shock. The source's older USD 55 billion2026 capex estimate is superseded by company guidance ofUSD 60-64 billion.

Of each incremental pricing dollar,85 cents becomes gross profit and 3 cents funds incremental operating expense. The remainder absorbs materials, fulfilment costs and concessions. This modest pricing treatment reflects customers' cost of waiting for qualified capacity without assuming that TSMC will maximize short-term scarcity rents. The October 8 capacity revision explicitly cancels the existing price benefit on incremental unserved N3 sales after deducting the old joint-loss overlap. This retains the original pricing probability and costs; scarcity does not create an automatic offsetting price increase.



Analytical model diagnostic | The forecast requires substantial leading-node capacity and accepted backend output; these quantities are model implications.

N3 capacity evidence and conditional delivery risk

The newly reviewed UBS Figure 13 estimates N3 installed capacity at 170/185/195/210 thousand wafers per month in 2027, against demand of 190/193/206/220. Its annual columns show 2.426 million demand wafers and 2.280 million nominal capacity wafers. Cloud AI accounts for 1.763 million, or 73% of demand. The crop has no report date and every quarter is an estimate. The source label “capacity utilization” means demand divided by capacity; it does not demonstrate saleable output above 100%.

The old model's USD 22,000 N3 wafer-price proxy implied approximately 283,000 monthly wafers in 2027. That is a different denominator from installed fab capacity. Applying 190,000/283,000 directly to sales would combine unverified scopes. Q2 reported N3 sales of NT$ 320,558.574 million, at 31.6/USD, equal USD 10,144.259 million. Matching those invoices to the image's 130,000 monthly capacity slots yields about USD 26,011 per nominal wafer slot. This normalizes one observed quarter. It is not a verified wafer ASP, yield estimate or physical output ceiling. Billing lag, inventory, customer mix and tool availability remain unresolved.

Holding that normalization constant gives conditional 2027 N3 sales of USD 13.27/14.44/15.22/16.39 billion, against the reference 15.58/17.71/19.84/21.52 billion. The model keeps the existing 30% joint execution probability. It revises that scenario's severity by taking the greater of the previous 2%/3% consolidated sales loss and the normalized N3 shortfall in each quarter. The 2027 conditional N3 gap is USD 15.34 billion; USD 4.54 billion was already included in the old joint loss. Only the difference changes expected sales, before linked CPU and price effects.

For 2028, the conditional scenario assumes 220/230/240/250 thousand monthly wafers, adding 10,000 per quarter from the 2027 exit. This 19% increase between year ends is an analyst assumption, not a UBS forecast or issuer commitment. It allows fabs, conversion and productivity to improve output while retaining the old 3% consolidated execution-loss floor. N5-to-N3 tool conversion cannot create capacity in both pools. The model adds no separate, unquantified N5 haircut on top of the combined loss.

Existing pricing is canceled only on incremental unserved N3 sales after deducting the old loss overlap. Incremental CPU capture is deferred by the N3 shortfall divided by reference N3 sales, bounded between 0% and 100%, within the same execution event. This is a workload-allocation proxy, not a claim that the whole CPU wallet uses N3. Its 60% capture probability, 65% gross margin and 8% operating costs remain. The 32 coherent states require both events for these interactions. Price or CPU cancellation occurs only where the corresponding upside event is present. Committed reference capex stays fixed; CPU incremental capex follows its deferred accepted sales.

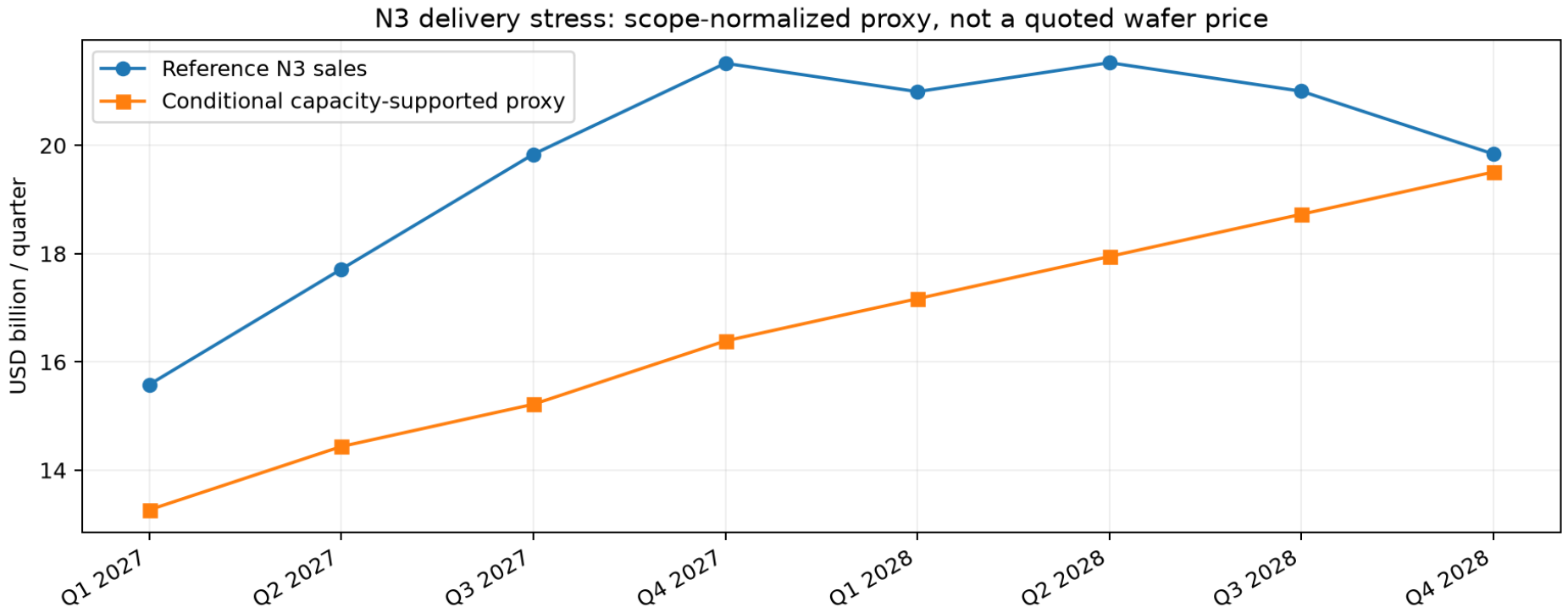
This lowers expected earnings while retaining the 20x valuation, Base 83, share schedule and opening financial claims. It is a testable capacity-risk update. A dated source, node shipment scope, qualified output, realized mix and tool-conversion evidence can resolve the remaining uncertainty. TSMC Q2 management transcript

Figure 13: N3 supply & demand analysis

Client	Products	Die size (mm2)	N3 capacity requirement (kwpm) Q126E	Q226E	Q326E	Q426E	2026 demand (k)	Q127E	Q227E	Q327E	Q427E	2027 demand (k)
Nvidia	Rubin	1,628	0	2	15	31	141	37	40	33	19	384
	Rubin Ultra	1,628	0	0	0	0	0	0	4	16	36	169
	Vera CPU	800	0	1	5	11	51	11	13	17	19	178
Broadcom	Google - TPU v7 Ironwood	1,400	6	11	16	21	158	14	8	6	0	85
	Google - TPU v8i	1,400	0	0	2	4	18	7	17	25	35	250
	Meta - MTIA v3	1,600	0	1	1	2	11	2	2	3	3	29
	OpenAI	1,600	0	0	0	3	10	3	3	3	3	32
AMD	MI355	840	2	2	1	1	17	1	1	0	0	7
Alchip	Amazon - Trainium 3	1,600	0	4	12	23	115	24	18	12	6	181
Marvell	Amazon - Trainium 2.5	1,600	0	0	0	2	6	0	0	0	0	2
MediaTek	Google - TPU v8t	800	0	0	1	4	15	9	11	11	13	129
Other CPUs / accelerators and products like networking			2	4	10	20	109	22	23	25	33	310
Capacity requirement - Cloud AI (kwpm)			**9**	**24**	**62**	**122**	**652**	**130**	**140**	**151**	**167**	**1,763**
% of N3 demand			***7%***	***18%***	***38%***	***64%***	***36%***	***69%***	***72%***	***73%***	***76%***	***73%***
Apple	iPhone's apps processor	110	43	43	39	22	442	13	11	9	10	128
	Mac's processor	165	10	10	8	7	106	7	6	6	6	74
	iPad's M processor for Pro & Air	160	14	12	12	8	135	11	10	8	8	111
	iPad's A17 Pro processor	104	2	2	1	1	15	3	2	2	2	30
Qualcomm	Flagship smartphone SoC	120	17	13	9	3	129	5	5	4	3	51
MediaTek	5G smartphone SoC	120	5	4	3	1	39	2	2	2	1	19
Intel	Outsourced PC CPUs	100	15	15	8	8	137	8	8	5	4	70
Others like Bitcoin, tablet, ARM based PC CPU			10	10	20	20	180	10	10	20	20	180
Capacity requirement - Consumer and other devices (kwpm)			**116**	**109**	**100**	**69**	**1,183**	**60**	**53**	**55**	**53**	**663**
% of N3 demand			***93%***	***82%***	***62%***	***36%***	***64%***	***31%***	***28%***	***27%***	***24%***	***27%***
Total N3 demand			**125**	**133**	**162**	**191**	**1,834**	**190**	**193**	**206**	**220**	**2,426**
TSMC's N3 installed capacity			**120**	**130**	**150**	**170**	**1,710**	**170**	**185**	**195**	**210**	**2,280**
Capacity utilisation			*104%*	*102%*	*108%*	*113%*	*107%*	*112%*	*104%*	*106%*	*105%*	*106%*

Source: Company data, UBS estimates. Note: kwpm stands for thousands of wafers per month.

User-supplied original UBS Figure 13 crop; source date unknown | Quarter columns are thousands of wafers per month; annual columns are thousands of wafers per year. Demand/capacity above 100% is an estimated load ratio, not demonstrated saleable output. The 2028 catch-up path is absent from this exhibit.



Analytical reconstruction from Q2 reported N3 sales and cropped UBS nominal capacity | The revenue gap is conditional on the calibrated capacity scope. It revises the existing 30% joint execution scenario after removing the prior loss overlap.

Gross profit and operating leverage

The model derives gross profit separately for each revenue vector. Those margins are allocations, not reported segment economics. Q3 node margins preserve their prior percentage-point differences through one common margin-point calibration to reported group GP after applying the actual wafer-mix percentages inside the retained non-wafer proxy. Q3 R&D/SG&A are issuer actuals. Q4 uses guidance midpoint GP and OP; its cost categories preserve the prior split. From 2027 onward, unchanged R&D combines a USD 2 billion quarterly program floor growing3.5% sequentially with 2.05% of sales; SG&A combines a USD 610 million floor growing2% sequentially with 0.8% of sales. Later node margins reflect ramp maturation, manufacturing efficiency and service mix. Incremental branches fund their own extra costs.

N2 ramp and overseas startup dilution are already embedded in the guided starting margin and the reference allocations. The adverse branch adds only costs beyond that baseline. At the unchanged 30% probability, the October 8 revision uses the greater of the previous 2%/3% loss floor and the normalized N3 shortfall in 2027-28. It applies70% lost gross-profit contribution and retains specified fixed/startup costs plus 0.3% extra operating expense. Later quarters keep the prior loss floor. Power delays, package qualification, yield and utilization problems form one joint stress rather than five independent demand shocks.

Capex is not operating expense at purchase. It becomes depreciation as assets enter service, while cash leaves through equipment payments. Ordinary depreciation is already inside the model's gross margin and operating costs; the funding check adds it back once and subtracts cash capex once. Q3 cash PP&E purchases of NT$ 443.206 billion and D&A of NT$ 220.204 billion demonstrate why the two cannot be substituted. Company-reported cash capex is USD 13.84 billion; the funding schedule separately translates exact NTD flows at the retained 32 convention.

COUPE and second order beneficiaries

COUPE integrates optical and electronic functions closer to compute and switching. The key manufacturing question is reliable, high-throughput assembly and test. TrendForce's October 7 public preview identifies fiber-array alignment as a constraint and describes3.2 Tbps and 6.4 Tbps roadmap stages. These rates are not TSMC revenues, and an optical-interconnect efficiency claim is not a whole-rack power reduction. TrendForce public preview

The model gives a35% probability to a small incremental service opportunity after a50% overlap deduction: conditional sales ofUSD 0.1/0.4/0.9/1.5 billion in 2027-30. Gross margin is 50%, incremental operating expense15%, and extra capex30% of revenue. There is no2026 increment because production readiness alone does not specify paid accepted units. The reference already contains substantial backend expansion.

Historical unmodeled supplier context, not renewed candidate admission: FitTech, Taiwan ticker6706, was identified in the October7 research. Its own FAS product supports optical assembly and passive/active alignment, linking a specific tool to the coupling bottleneck. Revenue depends on accepted machines, throughput, service and customer qualification. The counterweight is reported H1 EPS of negative TWD 2.85. Neither profitable scale nor pricing power follows automatically from the product's relevance. FitTech product TWSE profile

Global Unichip, Taiwan ticker3443, appeared in that historical context for advanced ASIC design and integration. Its issuer page confirms TSMC's35% ownership and sole-foundry relationship. The reported Tesla opportunity remains unverified. Separate NRE/IP earnings from low-margin turnkey wafer pass-through before modeling value. These historical notes assign no new Base, target or model. They are not re-promoted in this update: a dated bottleneck and credible enormous shareholder-upside bridge after costs, financing and dilution are required under the October11 admission policy. GUC company profile

ASML and Advantest are related supplier context. Qualified lithography and increasing test time per good chip can capture spending as foundry constraints are addressed. For ASML, system acceptance and service matter; for Advantest, test parallelism and customer program qualification matter. Customer spending, export restrictions and productivity improvements can reduce the expected benefit. Their current commercial order mix needs a separate review before any score or target. ASML results Advantest engineering update

Competition and thesis failure

Intel and Samsung can win through successful qualification, credible output and attractive customer economics. Samsung's July collaboration with Broadcom is evidence of a serious alternative, but its broad agreement scope includes memory and does not quantify lost TSMC wafer sales. Intel packaging can take backend work while helping customers ship chips still fabricated at TSMC. Packaging competition and wafer competition need separate bridges. Samsung announcement

The active model assigns25% probability to incremental competitive sales losses of 1% in 2028,2% in 2029 and 3% in 2030, with 65% lost gross-profit contribution and retention costs. Terafab negotiations produce no current contracted revenue in this model: control, scope, funding and qualified timing remain unsettled, with the cited development scenario extending beyond the modeled horizon. The exclusion is a timing and commercial-scope decision, not a dismissal of unofficial evidence.

The thesis weakens materially if customers reduce wafer commitments while capex remains fixed, if N2/N3 good output misses the modeled requirement, or if revenue growth comes with persistently lower economic returns. Additional failure signals include shrinking same-node pricing after concessions, outside manufacturing

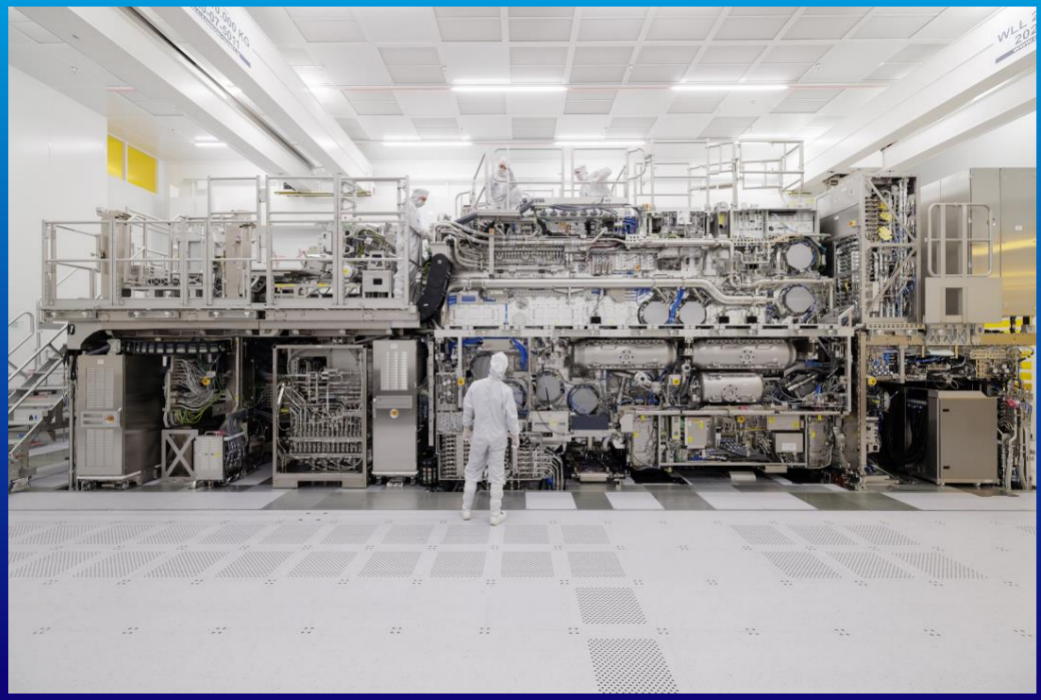
becoming competitive at commercial yield, and backend growth failing to convert to cash. No generic country discount is imposed; identifiable operating disruption belongs in costs, timing and output.

High NA EUV reaches new readiness milestone
with first high-volume Logic product

"This milestone reflects the close technical collaboration between Intel and ASML and shows how High NA EUV can be integrated into advanced semiconductor manufacturing at scale. By qualifying the High NA EUV process option on select Intel 18A product layers, our existing fleet of tools are providing customers with increased output, while we develop future options to achieve leading-edge performance, density and manufacturing flexibility on upcoming nodes."

Naga Chandrasekaran, Executive Vice President and General Manager of Intel Foundry.

intel



ASML July 15, 2026

Page 6
Public

Original supplier PDF page 6 | Intel High-NA readiness milestone, not proof of cross-foundry cost or yield parity.

Financial claims and the ADS denominator

June issued ordinary shares are 25,932.370 million. At five ordinary shares per ADS, the entire company has5,186.474 million ADS equivalents. The actual US receipt float is only a subset and must not divide consolidated earnings. The model adds0.1% annual net dilution and uses shares at the end of each four-quarter earnings window. The current active window uses 5,198.151 million ADS equivalents.

At the chosen NT$ 32/USD convention, June cash and current securities totalUSD 109.94 billion. The model reservesUSD 50 billion for operating needs, customer advances, near-term tax/dividend obligations and investment flexibility. Face debt, leases and pensions totalUSD 33.64 billion. This leaves a fixed net financial asset credit ofUSD 26.29 billion. No forecast retained cash is added again. Associates and long-term investment marks receive no separate asset credit in this operating-profit model.

September condensed statements round outstanding shares to 25,932 million, which does not replace the retained exact June issued-share anchor. Actual USD 9 million NCI net income is an after-tax amount, not an OP allocation. Outside owners receive a small rising modeled share of operating earnings:0.1% in 2026,0.2% in 2027,0.3% in 2028 and 0.4% later. This proxy is deliberately above the very small recent group NCI share as minority-owned overseas capacity ramps. Book NCI is not also deducted as debt. The Taiwan ordinary equivalent of the active ceiling is TWD 3,996.17 at the fixed exchange rate, before ADR premiums, fees or investor tax differences.

Cash funding and valuation sensitivities

The funding diagnostic now begins with reported September30 cash plus marketable instruments of NT$ 4,016,979 million, translated at the retained 32 convention to **USD 125.530594 billion**. This updates funding feasibility only; the valuation keeps its fixed opening net asset credit of USD 26.293434 billion. Higher cash is not added to the target a second time. The balance sheet's own USD convenience FX31.880 would give a different displayed USD cash amount.

Q3 reported CFO is NT$ 919,776 million, PP&E purchases443,206 million, dividends155,595 million and D&A220,204 million. Exact displayed CFO less PP&E is NT$ 476,570 million; the issuer management summary shows FCF476.56 billion after source rounding. At modelFX32, these flows are USD 28,743.0/13,850.1875/4,862.34375/6,881.375 million. The quarter's remaining liquidity movement is an explicit residual, including investment/tax-credit/FX and securities effects; it is not recurring OP. The NT$ 198.74 billion other investing inflow is mainly U.S. Treasury Investment Tax Credits. Do not classify it as customer sales.

The prior Q3 cash-capex assumption of USD 17.5 billion is replaced by the actual NTD flow at 32 only in the funding diagnostic. The USD 3.6498125 billion unspent difference moves into Q4, taking its cash-capex allowance to USD 21.3498125 billion. Retained H1 spending anchors USD 11.1/15.7 billion plus Q3 actual13.8501875 billion and the Q4 residual preserve the existing full-year2026 USD 62 billion funding plan. Company-reported YTD capex USD 40.64 billion implies about USD 21.36 billion nominal-USD spending to reach62; the small difference reflects translation/rounding scope. This is analyst payment timing, not a newly issued annual guide or a lower total investment plan. Reference2027-30 capex remains 80/94/106/112 billion; CPU/COUPE states fund additional capital separately. The previous issuer USD 60-64 billion budget remains historical guidance pending the new call; no new 2027 budget is verified. A separate USD 2 billion Sony-JV payment remains inQ1 2027 as a conservative sensitivity without invented JV revenue or asset credit.

Future cash generation keeps normal 19%taxes, a10%working-capital requirement on incremental sales, cash interest, depreciation addback, dividends and capex. All32 persistent combinations remain above the USD 50 billion reserve at modeled quarter ends; the minimum is USD 125.531 billion. This is a funding proxy with debt continuity, not a full maturity-by-maturity treasury forecast.

The retained USD 62.109 billion severe-payment diagnostic is an opening-obligation bound, not a newly disclosed September liability total. Subtracting it from updated September liquidity leaves USD 63.422 billion, above the unchanged USD 50 billion comparison floor. It retains possible double payments and does not prove intraperiod timing or alter the valuation's financial claims.

Catalysts and update discipline

The October15 actual/guidance anchor is incorporated. Next obtain a complete original earnings-call transcript or audio, especially any annual capex, AI/customer growth,2027 pricing, qualified N2/N3 output and packaging conversion disclosures. No Q3 transcript was linked on the reviewed official quarterly page; webcast and teleconference links led to registration. The retrieved release, presentation and six-page report establish no new quantitative2027 forecast or annual-capex guidance. A transcript access gap is not filled with a social relay.

The next monthly sales catalyst is October revenue, scheduled November10,2026; the call and future quarterly mix/qualified output provide the operational test. Every new price claim must identify node, repriceable contracts, date, realized costs and overlap. Aggregate ASP can change from mix.

Native reads for all four new issuer PDFs returned HTTP403. Supported official full text was read; requested screenshots returned tool errors or citation-only strings, not pixels. The chart-only technology/platform deck pages were cross-checked against the management report's complete text tables. No original-slide visual review is claimed. Prior WK/Morgan Stanley numerical-image and paid TrendForce limitations remain. These access limits do not prevent use of independently complete official tables.

Annual operating and funding crosscheck

USD million. All forecasts reconcile to the quarterly schedules. Non-wafer revenue includes advanced packaging, testing, masks and other services; the broader total is not pure CoWoS.

Year	Revenue	Gross profit	Operating expenses	Parent OP	Other revenue reference
2027	226,254.2	150,614.2	18,529.9	131,820.1	40,860.0
2028	290,686.9	193,604.8	22,194.8	170,895.8	57,600.0
2029	342,603.4	225,455.8	25,596.7	199,059.7	71,400.0
2030	380,629.4	251,503.1	28,834.8	221,777.6	79,800.0

The unchanged reference implies89,000/161,000 N2 monthly wafer equivalents and 283,000/316,000 N3 equivalents in 2027/28 at proxy ASPs. These are invoice equivalents, not reportedphysicalcapacity. TheOctober8 joint execution scenario tests scope-normalizedN3 capacity anddefers overlappingCPUoutput; an aggregatewafercountdoesnotprovequalifiedleading-nodeoutput.

Cash feasibility and extreme repayment test

The table uses the same operating forecast with no new net debt or equity. Starting principal is assumed refinanced in the normal path. All32 event combinations remain above the USD 50bn reserve at modeled quarter ends.

Quarter	Expected liquidity	Adverse liquidity	Cash capex	D&A addback	Dividends
Q3 2026	125,530.6	125,530.6	13,850.2	6,881.4	4,862.3
Q4 2026	130,597.3	130,597.3	21,349.8	8,100.0	5,500.0
Q1 2027	134,586.9	132,954.3	20,140.2	9,200.0	6,000.0
Q2 2027	142,970.2	139,326.2	20,138.6	9,900.0	6,000.0
Q3 2027	154,478.2	148,237.0	20,136.6	10,600.0	6,000.0
Q4 2027	169,300.2	160,121.3	20,136.3	11,300.0	6,000.0
Q1 2028	181,215.2	169,051.1	23,748.7	12,400.0	7,000.0
Q2 2028	196,305.7	181,126.7	23,749.9	13,400.0	7,000.0
Q3 2028	215,202.8	197,337.7	23,754.3	14,400.0	7,000.0
Q4 2028	237,583.7	216,993.0	23,761.2	15,400.0	7,000.0
Q1 2029	256,172.3	232,798.7	26,847.6	16,400.0	8,000.0
Q2 2029	277,508.5	251,252.2	26,847.6	17,400.0	8,000.0
Q3 2029	301,745.2	272,500.3	26,847.6	18,400.0	8,000.0
Q4 2029	329,096.7	296,760.8	26,847.6	19,400.0	8,000.0
Q1 2030	353,440.1	317,855.7	28,399.4	20,100.0	9,000.0
Q2 2030	380,219.2	341,284.6	28,399.4	20,800.0	9,000.0
Q3 2030	409,659.4	367,257.9	28,399.4	21,500.0	9,000.0
Q4 2030	441,884.1	395,900.6	28,399.4	22,200.0	9,000.0

The retained USD 62.109 billion opening-obligation severe-payment test leaves USD 63.422 billion from September cash plus securities, above the USD 50 billion floor. It is not a newly refreshed liability schedule or a forecast payment requirement; no debt or equity issuance is added.

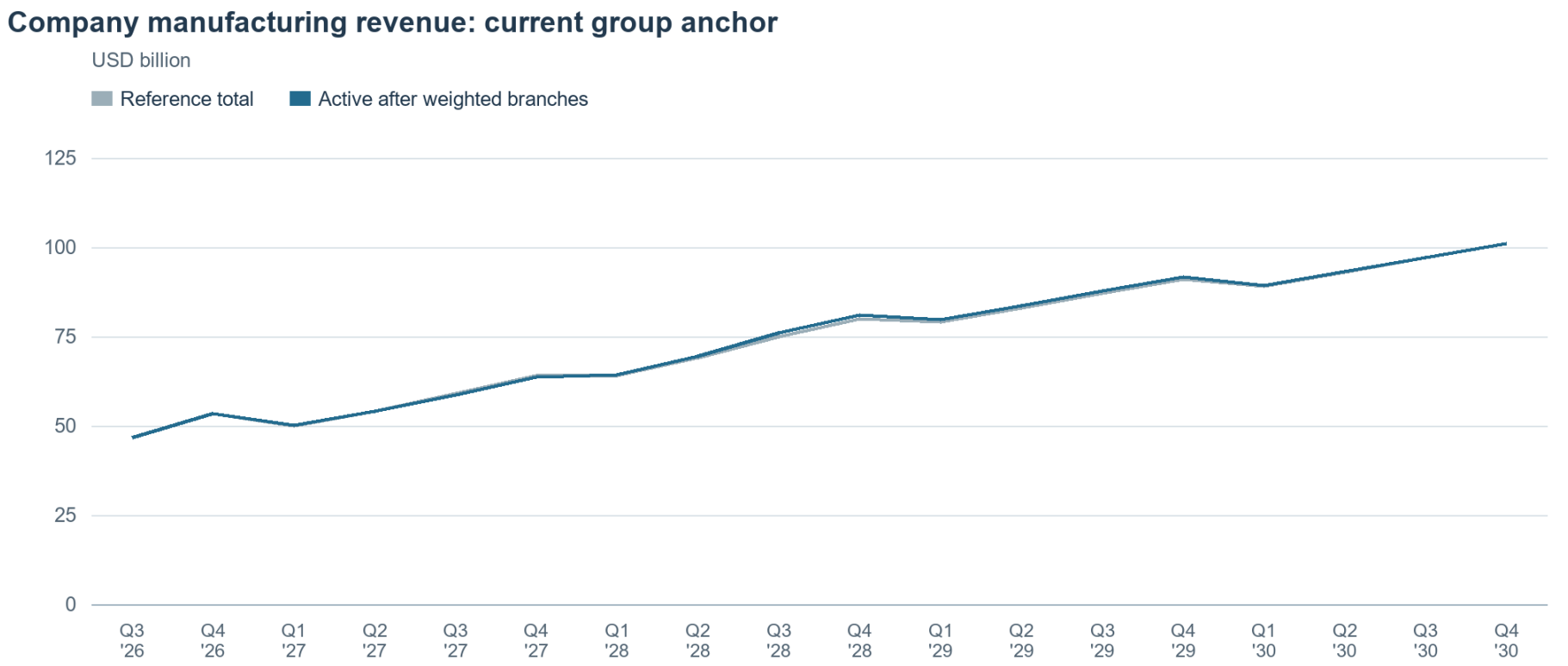
The table above weights the32coherent funding states and shows the minimum state separately. The linked Excel H/input path keeps the existing positive-working-capital MAX convention, so its cash path can differ slightly from weighted state cash after an upside/downside blend; it is separately named below and is not substituted for the state minimum. This pre-existing diagnostic distinction changes no OP, claims, shares or ceiling.

Quarter	State-weighted CFO	Other cash bridge	Reference capex	JV payment	Excel input-path liquidity
Q3 2026	28,743.0	5,562.2	13,850.2	0.0	125,530.6
Q4 2026	31,916.5	0.0	21,349.8	0.0	130,597.3
Q1 2027	32,129.8	0.0	20,000.0	2,000.0	134,586.9
Q2 2027	34,522.0	0.0	20,000.0	0.0	142,970.2
Q3 2027	37,644.6	0.0	20,000.0	0.0	154,478.2
Q4 2027	40,958.3	0.0	20,000.0	0.0	169,300.2
Q1 2028	42,663.7	0.0	23,500.0	0.0	181,218.0
Q2 2028	45,840.4	0.0	23,500.0	0.0	196,308.6
Q3 2028	49,651.4	0.0	23,500.0	0.0	215,205.7
Q4 2028	53,142.1	0.0	23,500.0	0.0	237,586.5
Q1 2029	53,436.3	0.0	26,500.0	0.0	256,175.2
Q2 2029	56,183.8	0.0	26,500.0	0.0	277,511.4
Q3 2029	59,084.3	0.0	26,500.0	0.0	301,748.1
Q4 2029	62,199.1	0.0	26,500.0	0.0	329,099.6
Q1 2030	61,742.8	0.0	28,000.0	0.0	353,443.0
Q2 2030	64,178.5	0.0	28,000.0	0.0	380,222.1
Q3 2030	66,839.5	0.0	28,000.0	0.0	409,662.3
Q4 2030	69,624.1	0.0	28,000.0	0.0	441,887.0

Score arithmetic

Category	Points	Evidence / treatment
Core thesis	19	Process/design ecosystem, commercial yield, wafer scale and packaging jointly create a hard-to-replicate qualified supply position.
Cycle	9	AI capacity demand strong into2027; consumers and power readiness limit straight-line extrapolation.
TAM	9	Accelerators, customsilicon, agenticCPUs andadvancedpackaging; wallet and overlap explicitly bounded.
Catalysts	4	Q3 actual and Q4 guide incorporated; full call/capex, N2/N3 qualified output and 2027 price realization remain tests.
Macro	9	Broad compute capital formation; neutralcountry baseline.
Alpha	3	The moat is widelyrecognized; alpha rests on earningsconversion andincrementalpricing,not discovery.
Risk classes	-3	Three classes: customer/cycleconcentration;manufacturing/overseasexecution;competitive/technology substitution. Cashfundingdoesnotadd anextra riskclass.
Total	50/60	Rounded Base 83. Artifact analytical Base 83 and existing workbook Base 83 remain unchanged; no new manual override is introduced.

Revenue and business mix

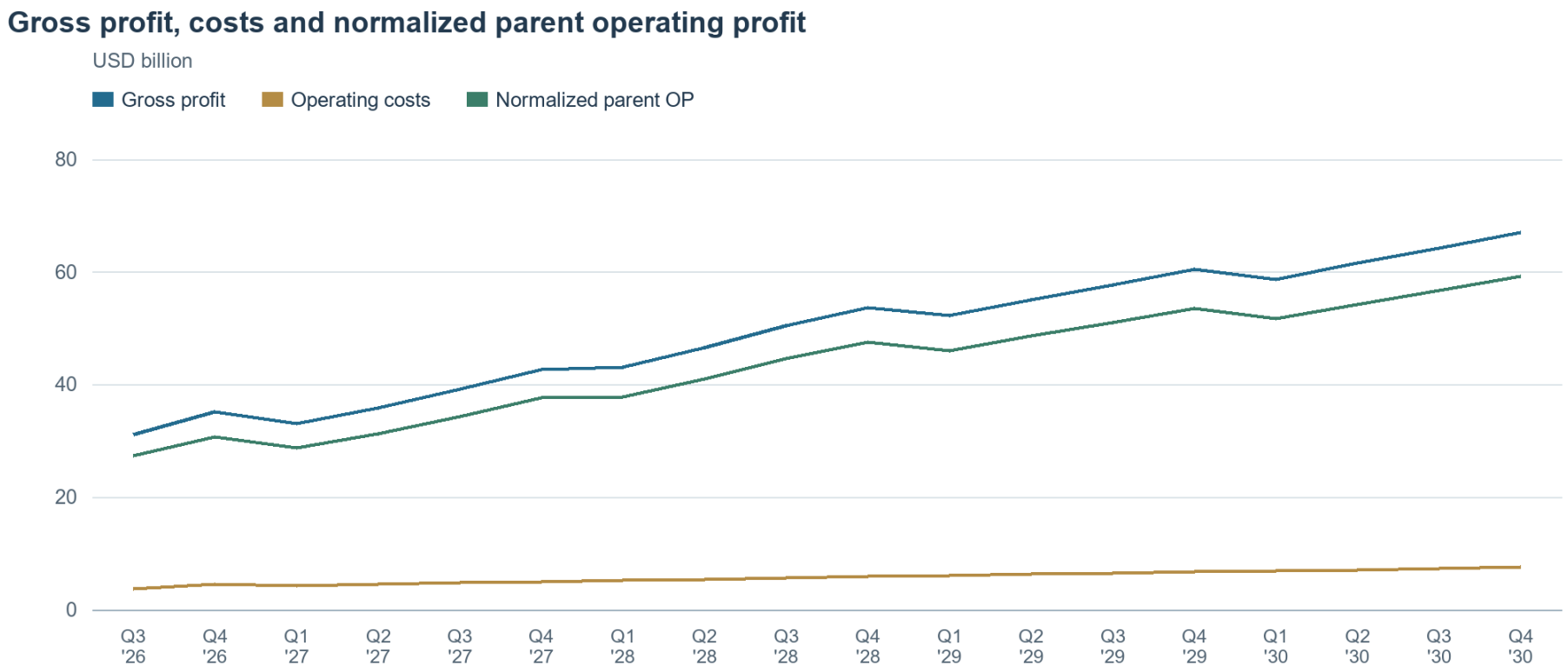


Q3 group revenue is issuer actual; Q4 is guidance midpoint. Product allocations are calibrated proxies and later rows are forecasts. The platform, node and backend views reconcile to one total.

Quarter	Reference revenue	Weighted net change	Active revenue
Q3 2026	46,720.0	0.0	46,720.0
Q4 2026	53,300.0	0.0	53,300.0
Q1 2027	50,000.0	181.1	50,181.1
Q2 2027	54,000.0	-50.0	53,950.0
Q3 2027	59,000.0	-394.1	58,605.9

Quarter	Reference revenue	Weighted net change	Active revenue
Q4 2027	64,000.0	-482.8	63,517.2
Q1 2028	64,000.0	345.4	64,345.4
Q2 2028	69,000.0	478.8	69,478.8
Q3 2028	75,000.0	930.0	75,930.0
Q4 2028	80,000.0	932.5	80,932.5
Q1 2029	79,000.0	734.3	79,734.3
Q2 2029	83,000.0	687.2	83,687.2
Q3 2029	87,000.0	624.3	87,624.3
Q4 2029	91,000.0	557.6	91,557.6
Q1 2030	89,000.0	274.1	89,274.1
Q2 2030	93,000.0	197.9	93,197.9
Q3 2030	97,000.0	119.3	97,119.3
Q4 2030	101,000.0	38.2	101,038.2

Gross profit, operating expenses and earnings

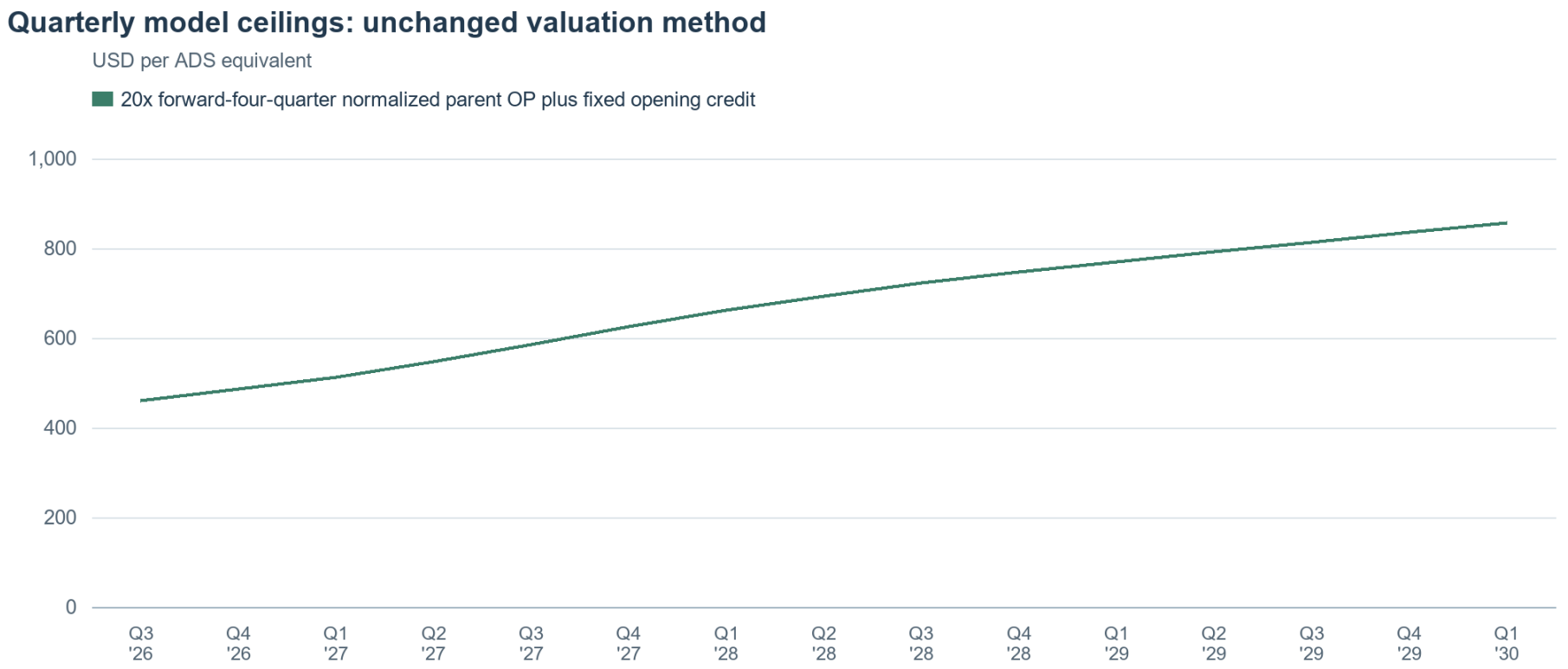


Q3 group GP/R&D/SG&A are actual; normalized parent OP excludes USD16M other operating income and applies the retained minority proxy. Q4 uses guidance midpoints; later quarters are forecasts.

Quarter	Revenue	GP	Opex	Valuation OP	OP margin
Q3 2026	46,720.0	31,028.0	3,641.0	27,359.6	58.6%
Q4 2026	53,300.0	35,178.0	4,530.5	30,616.9	57.4%
Q1 2027	50,181.1	33,037.8	4,297.3	28,683.0	57.2%
Q2 2027	53,950.0	35,799.0	4,504.1	31,232.3	57.9%
Q3 2027	58,605.9	39,138.3	4,743.1	34,326.5	58.6%

Quarter	Revenue	GP	Opex	Valuation OP	OP margin
Q4 2027	63,517.2	42,639.1	4,985.5	37,578.3	59.2%
Q1 2028	64,345.4	43,053.1	5,152.6	37,786.8	58.7%
Q2 2028	69,478.8	46,520.0	5,404.2	40,992.4	59.0%
Q3 2028	75,930.0	50,438.5	5,689.6	44,614.6	58.8%
Q4 2028	80,932.5	53,593.3	5,948.4	47,502.0	58.7%
Q1 2029	79,734.3	52,201.5	6,045.5	45,971.3	57.7%
Q2 2029	83,687.2	55,080.4	6,279.0	48,606.2	58.1%
Q3 2029	87,624.3	57,661.5	6,515.8	50,941.2	58.1%
Q4 2029	91,557.6	60,512.4	6,756.3	53,541.0	58.5%
Q1 2030	89,274.1	58,674.4	6,831.4	51,635.6	57.8%
Q2 2030	93,197.9	61,550.3	7,080.1	54,252.3	58.2%
Q3 2030	97,119.3	64,224.0	7,333.0	56,663.4	58.3%
Q4 2030	101,038.2	67,054.4	7,590.3	59,226.3	58.6%

The complete ceiling path



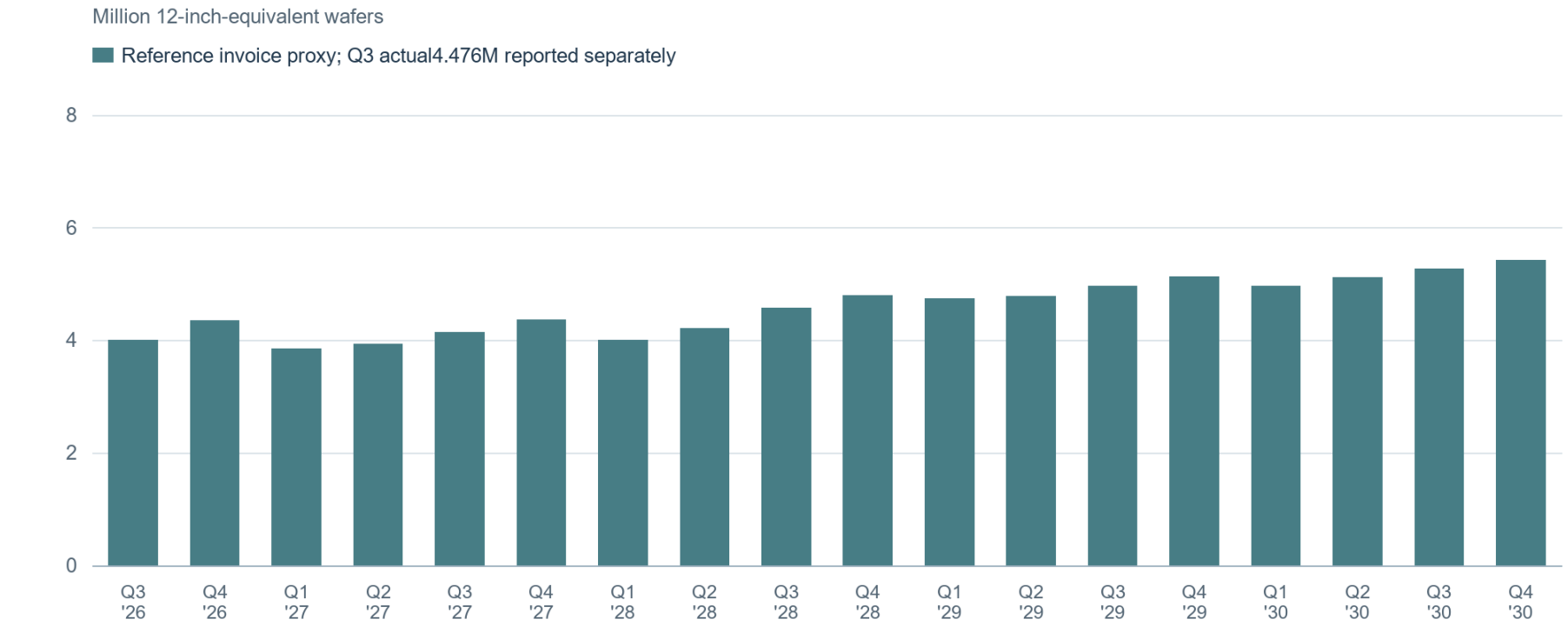
Quarter Q values Q plus the next three quarters. Q3/Q4 2026 ceilings change; the active Q4 2027 ceiling remains USD624.4020697630581. Fixed opening credit and end-window share convention remain.

Quarter	Four-quarter earnings window	F4Q OP	Multiple	Shares	Ceiling
Q3 2026	Q3 2026 to Q2 2027	117,891.8	20x	5,191.660	459.22
Q4 2026	Q4 2026 to Q3 2027	124,858.7	20x	5,192.958	485.94
Q1 2027	Q1 2027 to Q4 2027	131,820.1	20x	5,194.256	512.62
Q2 2027	Q2 2027 to Q1 2028	140,923.9	20x	5,195.554	547.54

Quarter	Four-quarter earnings window	F4Q OP	Multiple	Shares	Ceiling
Q3 2027	Q3 2027 to Q2 2028	150,684.0	20x	5,196.852	584.96
Q4 2027	Q4 2027 to Q3 2028	160,972.1	20x	5,198.151	624.40
Q1 2028	Q1 2028 to Q4 2028	170,895.8	20x	5,199.450	662.42
Q2 2028	Q2 2028 to Q1 2029	179,080.3	20x	5,200.749	693.73
Q3 2028	Q3 2028 to Q2 2029	186,694.0	20x	5,202.049	722.83
Q4 2028	Q4 2028 to Q3 2029	193,020.6	20x	5,203.349	746.96
Q1 2029	Q1 2029 to Q4 2029	199,059.7	20x	5,204.649	769.98
Q2 2029	Q2 2029 to Q1 2030	204,724.0	20x	5,205.950	791.55
Q3 2029	Q3 2029 to Q2 2030	210,370.1	20x	5,207.251	813.04
Q4 2029	Q4 2029 to Q3 2030	216,092.4	20x	5,208.552	834.81
Q1 2030	Q1 2030 to Q4 2030	221,777.6	20x	5,209.854	856.42

The operating constraint behind the forecast

Wafer invoice equivalents: analytical scope, not qualified capacity



The reference invoice proxy converts modeled node sales using stated ASP proxies. Actual Q3 shipments4.476M are a separate issuer fact. An aggregate count does not prove N3 qualified capacity or customer-accepted units.

Quarter	Reference wafer equivalents M (Q3 actual shown separately)	Status
Q3 2026	4.007	Issuer actual shipments4.476M; model invoice proxy and qualified-node capacity are different scopes.
Q4 2026	4.341	Guidance-calibrated analytical invoice proxy
Q1 2027	3.848	Forecast assumption
Q2 2027	3.929	Forecast assumption

Quarter	Reference wafer equivalents M (Q3 actual shown separately)	Status
Q3 2027	4.145	Forecast assumption
Q4 2027	4.355	Forecast assumption
Q1 2028	3.998	Forecast assumption
Q2 2028	4.204	Forecast assumption
Q3 2028	4.568	Forecast assumption
Q4 2028	4.798	Forecast assumption
Q1 2029	4.743	Forecast assumption
Q2 2029	4.785	Forecast assumption
Q3 2029	4.958	Forecast assumption
Q4 2029	5.125	Forecast assumption
Q1 2030	4.953	Forecast assumption
Q2 2030	5.114	Forecast assumption
Q3 2030	5.269	Forecast assumption
Q4 2030	5.419	Forecast assumption

Quarterly revenue-vector derivation

All rows below are reference inputs. Gross profit = vector revenue x the stated gross margin, except explicitly labeled zero-sales cost credits. The next table applies the separately weighted evidence deltas. A product-level proxy is a model assumption even when the group total is calibrated to guidance or preliminary monthly sales. Q3 wafer technology shares are reported percentages on the wafer denominator; non-wafer dollar share and individual vector gross margins remain calibrated proxies. Actual total shipments are reported separately.

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2026	N2 / A16 wafer fabrication	3,139.6	50.4%	1,583.6
Q3 2026	N3 wafer fabrication	12,950.8	76.4%	9,899.4
Q3 2026	N5 / N7 wafer fabrication	14,913.0	73.9%	11,026.5

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2026	Other mature and specialty wafers	8,241.4	54.4%	4,486.5
Q3 2026	A14 / A13 / A12 wafer fabrication	0.0	0.0%	0.0
Q3 2026	Advanced packaging, testing and other non-wafer services	7,475.2	53.9%	4,032.0
Q4 2026	N2 / A16 wafer fabrication	5,785.7	49.6%	2,870.7
Q4 2026	N3 wafer fabrication	15,131.9	77.1%	11,669.2
Q4 2026	N5 / N7 wafer fabrication	14,686.8	74.1%	10,885.4
Q4 2026	Other mature and specialty wafers	8,901.1	54.6%	4,861.5
Q4 2026	A14 / A13 / A12 wafer fabrication	0.0	0.0%	0.0
Q4 2026	Advanced packaging, testing and other non-wafer services	8,794.5	55.6%	4,891.2
Q1 2027	N2 / A16 wafer fabrication	5,740.0	53.0%	3,042.2
Q1 2027	N3 wafer fabrication	15,580.0	77.0%	11,996.6
Q1 2027	N5 / N7 wafer fabrication	11,890.0	73.5%	8,739.1
Q1 2027	Other mature and specialty wafers	7,790.0	54.0%	4,206.6
Q1 2027	A14 / A13 / A12 wafer fabrication	0.0	0.0%	0.0
Q1 2027	Advanced packaging, testing and other non-wafer services	9,000.0	56.0%	5,040.0
Q2 2027	N2 / A16 wafer fabrication	7,084.8	56.0%	3,967.5
Q2 2027	N3 wafer fabrication	17,712.0	77.0%	13,638.2
Q2 2027	N5 / N7 wafer fabrication	11,955.6	73.5%	8,787.4
Q2 2027	Other mature and specialty wafers	7,527.6	54.0%	4,064.9

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q2 2027	A14 / A13 / A12 wafer fabrication	0.0	0.0%	0.0
Q2 2027	Advanced packaging, testing and other non-wafer services	9,720.0	56.5%	5,491.8
Q3 2027	N2 / A16 wafer fabrication	8,708.4	59.0%	5,138.0
Q3 2027	N3 wafer fabrication	19,835.8	77.0%	15,273.6
Q3 2027	N5 / N7 wafer fabrication	12,095.0	73.5%	8,889.8
Q3 2027	Other mature and specialty wafers	7,740.8	54.0%	4,180.0
Q3 2027	A14 / A13 / A12 wafer fabrication	0.0	0.0%	0.0
Q3 2027	Advanced packaging, testing and other non-wafer services	10,620.0	57.0%	6,053.4
Q4 2027	N2 / A16 wafer fabrication	10,496.0	61.0%	6,402.6
Q4 2027	N3 wafer fabrication	21,516.8	77.0%	16,567.9
Q4 2027	N5 / N7 wafer fabrication	12,595.2	73.5%	9,257.5
Q4 2027	Other mature and specialty wafers	7,872.0	54.0%	4,250.9
Q4 2027	A14 / A13 / A12 wafer fabrication	0.0	0.0%	0.0
Q4 2027	Advanced packaging, testing and other non-wafer services	11,520.0	57.5%	6,624.0
Q1 2028	N2 / A16 wafer fabrication	11,264.0	63.0%	7,096.3
Q1 2028	N3 wafer fabrication	20,992.0	76.5%	16,058.9
Q1 2028	N5 / N7 wafer fabrication	11,776.0	73.0%	8,596.5
Q1 2028	Other mature and specialty wafers	6,656.0	53.0%	3,527.7
Q1 2028	A14 / A13 / A12 wafer fabrication	512.0	42.0%	215.0

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q1 2028	Advanced packaging, testing and other non-wafer services	12,800.0	58.0%	7,424.0
Q2 2028	N2 / A16 wafer fabrication	13,248.0	65.0%	8,611.2
Q2 2028	N3 wafer fabrication	21,528.0	76.5%	16,468.9
Q2 2028	N5 / N7 wafer fabrication	12,144.0	73.0%	8,865.1
Q2 2028	Other mature and specialty wafers	6,900.0	53.0%	3,657.0
Q2 2028	A14 / A13 / A12 wafer fabrication	1,380.0	45.0%	621.0
Q2 2028	Advanced packaging, testing and other non-wafer services	13,800.0	58.5%	8,073.0
Q3 2028	N2 / A16 wafer fabrication	15,600.0	66.0%	10,296.0
Q3 2028	N3 wafer fabrication	21,000.0	76.0%	15,960.0
Q3 2028	N5 / N7 wafer fabrication	12,600.0	73.0%	9,198.0
Q3 2028	Other mature and specialty wafers	7,800.0	53.0%	4,134.0
Q3 2028	A14 / A13 / A12 wafer fabrication	3,000.0	49.0%	1,470.0
Q3 2028	Advanced packaging, testing and other non-wafer services	15,000.0	59.0%	8,850.0
Q4 2028	N2 / A16 wafer fabrication	17,920.0	67.0%	12,006.4
Q4 2028	N3 wafer fabrication	19,840.0	76.0%	15,078.4
Q4 2028	N5 / N7 wafer fabrication	12,800.0	73.0%	9,344.0
Q4 2028	Other mature and specialty wafers	8,320.0	53.0%	4,409.6
Q4 2028	A14 / A13 / A12 wafer fabrication	5,120.0	53.0%	2,713.6
Q4 2028	Advanced packaging, testing and other non-wafer services	16,000.0	59.5%	9,520.0

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q1 2029	N2 / A16 wafer fabrication	18,098.9	68.0%	12,307.3
Q1 2029	N3 wafer fabrication	16,850.7	75.0%	12,638.0
Q1 2029	N5 / N7 wafer fabrication	11,857.9	72.0%	8,537.7
Q1 2029	Other mature and specialty wafers	8,737.4	52.0%	4,543.4
Q1 2029	A14 / A13 / A12 wafer fabrication	6,865.1	56.0%	3,844.5
Q1 2029	Advanced packaging, testing and other non-wafer services	16,590.0	60.0%	9,954.0
Q2 2029	N2 / A16 wafer fabrication	18,359.6	69.0%	12,668.1
Q2 2029	N3 wafer fabrication	17,048.2	75.0%	12,786.2
Q2 2029	N5 / N7 wafer fabrication	11,802.6	72.0%	8,497.9
Q2 2029	Other mature and specialty wafers	8,524.1	52.0%	4,432.5
Q2 2029	A14 / A13 / A12 wafer fabrication	9,835.5	60.0%	5,901.3
Q2 2029	Advanced packaging, testing and other non-wafer services	17,430.0	60.0%	10,458.0
Q3 2029	N2 / A16 wafer fabrication	17,869.8	69.0%	12,330.2
Q3 2029	N3 wafer fabrication	17,182.5	74.5%	12,801.0
Q3 2029	N5 / N7 wafer fabrication	11,684.1	72.0%	8,412.6
Q3 2029	Other mature and specialty wafers	8,934.9	52.0%	4,646.1
Q3 2029	A14 / A13 / A12 wafer fabrication	13,058.7	63.0%	8,227.0
Q3 2029	Advanced packaging, testing and other non-wafer services	18,270.0	60.0%	10,962.0
Q4 2029	N2 / A16 wafer fabrication	17,253.6	70.0%	12,077.5

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q4 2029	N3 wafer fabrication	17,253.6	74.5%	12,853.9
Q4 2029	N5 / N7 wafer fabrication	11,502.4	72.0%	8,281.7
Q4 2029	Other mature and specialty wafers	9,345.7	52.0%	4,859.8
Q4 2029	A14 / A13 / A12 wafer fabrication	16,534.7	65.0%	10,747.6
Q4 2029	Advanced packaging, testing and other non-wafer services	19,110.0	60.0%	11,466.0
Q1 2030	N2 / A16 wafer fabrication	15,468.2	70.0%	10,827.7
Q1 2030	N3 wafer fabrication	16,171.3	74.0%	11,966.8
Q1 2030	N5 / N7 wafer fabrication	10,546.5	71.0%	7,488.0
Q1 2030	Other mature and specialty wafers	9,140.3	51.0%	4,661.6
Q1 2030	A14 / A13 / A12 wafer fabrication	18,983.7	66.0%	12,529.2
Q1 2030	Advanced packaging, testing and other non-wafer services	18,690.0	60.0%	11,214.0
Q2 2030	N2 / A16 wafer fabrication	14,694.0	70.0%	10,285.8
Q2 2030	N3 wafer fabrication	16,163.4	74.0%	11,960.9
Q2 2030	N5 / N7 wafer fabrication	10,285.8	71.0%	7,302.9
Q2 2030	Other mature and specialty wafers	9,551.1	51.0%	4,871.1
Q2 2030	A14 / A13 / A12 wafer fabrication	22,775.7	68.0%	15,487.5
Q2 2030	Advanced packaging, testing and other non-wafer services	19,530.0	60.0%	11,718.0
Q3 2030	N2 / A16 wafer fabrication	13,793.4	70.0%	9,655.4
Q3 2030	N3 wafer fabrication	16,092.3	73.5%	11,827.8

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2030	N5 / N7 wafer fabrication	9,961.9	71.0%	7,072.9
Q3 2030	Other mature and specialty wafers	9,961.9	51.0%	5,080.6
Q3 2030	A14 / A13 / A12 wafer fabrication	26,820.5	69.0%	18,506.1
Q3 2030	Advanced packaging, testing and other non-wafer services	20,370.0	60.0%	12,222.0
Q4 2030	N2 / A16 wafer fabrication	12,766.4	70.0%	8,936.5
Q4 2030	N3 wafer fabrication	15,958.0	73.5%	11,729.1
Q4 2030	N5 / N7 wafer fabrication	9,574.8	71.0%	6,798.1
Q4 2030	Other mature and specialty wafers	10,372.7	51.0%	5,290.1
Q4 2030	A14 / A13 / A12 wafer fabrication	31,118.1	70.0%	21,782.7
Q4 2030	Advanced packaging, testing and other non-wafer services	21,210.0	60.0%	12,726.0

Operating-cost derivation

Quarter	R&D incl ordinary amortization and awards	SG&A incl ordinary depreciation and awards
Q3 2026	2,782.0	859.0
Q4 2026	3,404.5	1,126.0
Q1 2027	3,167.4	1,034.6
Q2 2027	3,324.4	1,079.3
Q3 2027	3,504.5	1,132.3
Q4 2027	3,687.4	1,185.5
Q1 2028	3,770.5	1,199.0
Q2 2028	3,959.1	1,252.7
Q3 2028	4,171.1	1,314.7
Q4 2028	4,365.8	1,369.0
Q1 2029	4,440.7	1,375.6
Q2 2029	4,621.4	1,422.5
Q3 2029	4,805.6	1,469.6
Q4 2029	4,993.4	1,517.1
Q1 2030	5,061.9	1,516.9
Q2 2030	5,257.2	1,565.0
Q3 2030	5,456.5	1,613.4
Q4 2030	5,659.9	1,662.1

The categories reconcile to reference operating expenses. Incremental costs below are applied once; depreciation or stock compensation already included in a reference category is not added again.

Probability-weighted financial bridge

Active = reference + sum(probability x bounded conditional delta). Negative deltas are retained. A branch with zero near-term increment remains zero only for the timing or overlap reasons documented in the evidence register.

Weighted OP below is after the stated minority operating-profit attribution. Conditional gross profit and expenses remain consolidated figures.

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2026	PRICE	65%	0.0	0.0	0.0	0.0
Q3 2026	CPU CAPTURE	60%	0.0	0.0	0.0	0.0
Q3 2026	COUPE OPTION	35%	0.0	0.0	0.0	0.0
Q3 2026	EXECUTION	30%	0.0	0.0	0.0	0.0
Q3 2026	COMPETITION	25%	0.0	0.0	0.0	0.0
Q4 2026	PRICE	65%	0.0	0.0	0.0	0.0
Q4 2026	CPU CAPTURE	60%	0.0	0.0	0.0	0.0
Q4 2026	COUPE OPTION	35%	0.0	0.0	0.0	0.0
Q4 2026	EXECUTION	30%	0.0	0.0	0.0	0.0
Q4 2026	COMPETITION	25%	0.0	0.0	0.0	0.0
Q1 2027	PRICE	65%	639.6	543.7	19.2	340.2
Q1 2027	CPU CAPTURE	60%	800.0	520.0	64.0	273.1
Q1 2027	COUPE OPTION	35%	25.0	12.5	3.8	3.1
Q1 2027	EXECUTION	30%	-2,411.4	-2,188.2	143.5	-698.1
Q1 2027	COMPETITION	25%	0.0	0.0	0.0	0.0
Q2 2027	PRICE	65%	743.9	632.3	22.3	395.7
Q2 2027	CPU CAPTURE	60%	800.0	520.0	64.0	273.1
Q2 2027	COUPE OPTION	35%	25.0	12.5	3.8	3.1
Q2 2027	EXECUTION	30%	-3,407.5	-2,927.3	153.6	-922.4
Q2 2027	COMPETITION	25%	0.0	0.0	0.0	0.0
Q3 2027	PRICE	65%	856.3	727.9	25.7	455.5
Q3 2027	CPU CAPTURE	60%	800.0	520.0	64.0	273.1
Q3 2027	COUPE OPTION	35%	25.0	12.5	3.8	3.1
Q3 2027	EXECUTION	30%	-4,798.3	-3,953.3	166.0	-1,233.3

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2027	COMPETITION	25%	0.0	0.0	0.0	0.0
Q4 2027	PRICE	65%	960.4	816.3	28.8	510.9
Q4 2027	CPU CAPTURE	60%	800.0	520.0	64.0	273.1
Q4 2027	COUPE OPTION	35%	25.0	12.5	3.8	3.1
Q4 2027	EXECUTION	30%	-5,319.4	-4,369.1	180.6	-1,362.2
Q4 2027	COMPETITION	25%	0.0	0.0	0.0	0.0
Q1 2028	PRICE	65%	1,290.2	1,096.7	38.7	685.6
Q1 2028	CPU CAPTURE	60%	1,400.0	910.0	112.0	477.4
Q1 2028	COUPE OPTION	35%	100.0	50.0	15.0	12.2
Q1 2028	EXECUTION	30%	-4,027.4	-3,458.9	178.3	-1,087.9
Q1 2028	COMPETITION	25%	-640.0	-416.0	128.0	-135.6
Q2 2028	PRICE	65%	1,391.0	1,182.4	41.7	739.2
Q2 2028	CPU CAPTURE	60%	1,400.0	910.0	112.0	477.4
Q2 2028	COUPE OPTION	35%	100.0	50.0	15.0	12.2
Q2 2028	EXECUTION	30%	-3,759.4	-3,320.5	194.6	-1,051.4
Q2 2028	COMPETITION	25%	-690.0	-448.5	138.0	-146.2
Q3 2028	PRICE	65%	1,464.0	1,244.4	43.9	778.0
Q3 2028	CPU CAPTURE	60%	1,400.0	910.0	112.0	477.4
Q3 2028	COUPE OPTION	35%	100.0	50.0	15.0	12.2
Q3 2028	EXECUTION	30%	-2,363.6	-2,400.1	217.7	-783.0
Q3 2028	COMPETITION	25%	-750.0	-487.5	150.0	-158.9
Q4 2028	PRICE	65%	1,510.4	1,283.8	45.3	802.6
Q4 2028	CPU CAPTURE	60%	1,400.0	910.0	112.0	477.4
Q4 2028	COUPE OPTION	35%	100.0	50.0	15.0	12.2
Q4 2028	EXECUTION	30%	-2,414.0	-2,489.1	238.9	-815.9

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q4 2028	COMPETITION	25%	-800.0	-520.0	160.0	-169.5
Q1 2029	PRICE	65%	1,048.5	891.2	31.5	556.6
Q1 2029	CPU CAPTURE	60%	1,800.0	1,170.0	144.0	613.1
Q1 2029	COUPE OPTION	35%	225.0	112.5	33.8	27.5
Q1 2029	EXECUTION	30%	-2,370.0	-2,291.0	237.0	-755.4
Q1 2029	COMPETITION	25%	-1,580.0	-1,027.0	158.0	-295.1
Q2 2029	PRICE	65%	1,062.2	902.9	31.9	563.9
Q2 2029	CPU CAPTURE	60%	1,800.0	1,170.0	144.0	613.1
Q2 2029	COUPE OPTION	35%	225.0	112.5	33.8	27.5
Q2 2029	EXECUTION	30%	-2,490.0	-2,407.0	249.0	-793.6
Q2 2029	COMPETITION	25%	-1,660.0	-1,079.0	166.0	-310.0
Q3 2029	PRICE	65%	1,051.6	893.8	31.5	558.2
Q3 2029	CPU CAPTURE	60%	1,800.0	1,170.0	144.0	613.1
Q3 2029	COUPE OPTION	35%	225.0	112.5	33.8	27.5
Q3 2029	EXECUTION	30%	-2,610.0	-2,523.0	261.0	-831.9
Q3 2029	COMPETITION	25%	-1,740.0	-1,131.0	174.0	-324.9
Q4 2029	PRICE	65%	1,035.2	879.9	31.1	549.6
Q4 2029	CPU CAPTURE	60%	1,800.0	1,170.0	144.0	613.1
Q4 2029	COUPE OPTION	35%	225.0	112.5	33.8	27.5
Q4 2029	EXECUTION	30%	-2,730.0	-2,639.0	273.0	-870.1
Q4 2029	COMPETITION	25%	-1,820.0	-1,183.0	182.0	-339.9
Q1 2030	PRICE	65%	632.8	537.9	19.0	335.9
Q1 2030	CPU CAPTURE	60%	2,000.0	1,300.0	160.0	681.3
Q1 2030	COUPE OPTION	35%	375.0	187.5	56.2	45.8
Q1 2030	EXECUTION	30%	-2,670.0	-2,581.0	267.0	-851.0

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q1 2030	COMPETITION	25%	-2,670.0	-1,735.5	178.0	-476.5
Q2 2030	PRICE	65%	617.1	524.6	18.5	327.6
Q2 2030	CPU CAPTURE	60%	2,000.0	1,300.0	160.0	681.3
Q2 2030	COUPE OPTION	35%	375.0	187.5	56.2	45.8
Q2 2030	EXECUTION	30%	-2,790.0	-2,697.0	279.0	-889.2
Q2 2030	COMPETITION	25%	-2,790.0	-1,813.5	186.0	-497.9
Q3 2030	PRICE	65%	597.7	508.1	17.9	317.3
Q3 2030	CPU CAPTURE	60%	2,000.0	1,300.0	160.0	681.3
Q3 2030	COUPE OPTION	35%	375.0	187.5	56.2	45.8
Q3 2030	EXECUTION	30%	-2,910.0	-2,813.0	291.0	-927.5
Q3 2030	COMPETITION	25%	-2,910.0	-1,891.5	194.0	-519.3
Q4 2030	PRICE	65%	574.5	488.3	17.2	305.0
Q4 2030	CPU CAPTURE	60%	2,000.0	1,300.0	160.0	681.3
Q4 2030	COUPE OPTION	35%	375.0	187.5	56.2	45.8
Q4 2030	EXECUTION	30%	-3,030.0	-2,929.0	303.0	-965.7
Q4 2030	COMPETITION	25%	-3,030.0	-1,969.5	202.0	-540.7

Consolidated profit to the common shareholder claim

Minority attribution is an earnings allocation, not operating expense or a fixed book-value deduction. Its rate is a disclosed model proxy and moves with reference profit and each probability-weighted change.

Quarter	Normalized consolidated OP	Minority rate proxy	Minority allocation	Owner OP
Q3 2026	27,387.0	0.1%	27.4	27,359.6
Q4 2026	30,647.5	0.1%	30.6	30,616.9
Q1 2027	28,740.5	0.2%	57.5	28,683.0
Q2 2027	31,294.9	0.2%	62.6	31,232.3

Quarter	Normalized consolidated OP	Minority rate proxy	Minority allocation	Owner OP
Q3 2027	34,395.2	0.2%	68.8	34,326.5
Q4 2027	37,653.6	0.2%	75.3	37,578.3
Q1 2028	37,900.5	0.3%	113.7	37,786.8
Q2 2028	41,115.8	0.3%	123.3	40,992.4
Q3 2028	44,748.8	0.3%	134.2	44,614.6
Q4 2028	47,644.9	0.3%	142.9	47,502.0
Q1 2029	46,155.9	0.4%	184.6	45,971.3
Q2 2029	48,801.4	0.4%	195.2	48,606.2
Q3 2029	51,145.7	0.4%	204.6	50,941.2
Q4 2029	53,756.1	0.4%	215.0	53,541.0
Q1 2030	51,843.0	0.4%	207.4	51,635.6
Q2 2030	54,470.2	0.4%	217.9	54,252.3
Q3 2030	56,891.0	0.4%	227.6	56,663.4
Q4 2030	59,464.1	0.4%	237.9	59,226.3

Enterprise-to-equity reconciliation

20x forward-four-quarter normalized IFRS operating profit attributable to parent, plus fixed opening surplus liquidity less face debt, lease and pension claims, divided by end-window ADS-equivalent shares. Each ADS represents five ordinary shares; Q prices Q plus the next three quarters.

Valuation quarter	Claims convention date	Reference claims	Weighted change	Total claims used	Ceiling / share
Q3 2026	Q2 2027	-26,293.4	0.0	-26,293.4	459.22
Q4 2026	Q3 2027	-26,293.4	0.0	-26,293.4	485.94
Q1 2027	Q4 2027	-26,293.4	0.0	-26,293.4	512.62
Q2 2027	Q1 2028	-26,293.4	0.0	-26,293.4	547.54
Q3 2027	Q2 2028	-26,293.4	0.0	-26,293.4	584.96
Q4 2027	Q3 2028	-26,293.4	0.0	-26,293.4	624.40

Valuation quarter	Claims convention date	Reference claims	Weighted change	Total claims used	Ceiling / share
Q1 2028	Q4 2028	-26,293.4	0.0	-26,293.4	662.42
Q2 2028	Q1 2029	-26,293.4	0.0	-26,293.4	693.73
Q3 2028	Q2 2029	-26,293.4	0.0	-26,293.4	722.83
Q4 2028	Q3 2029	-26,293.4	0.0	-26,293.4	746.96
Q1 2029	Q4 2029	-26,293.4	0.0	-26,293.4	769.98
Q2 2029	Q1 2030	-26,293.4	0.0	-26,293.4	791.55
Q3 2029	Q2 2030	-26,293.4	0.0	-26,293.4	813.04
Q4 2029	Q3 2030	-26,293.4	0.0	-26,293.4	834.81
Q1 2030	Q4 2030	-26,293.4	0.0	-26,293.4	856.42

The displayed convention date is the end of each earnings window, while the valuation financial-asset credit is deliberately fixed at the original opening balance. The separate funding schedule is refreshed to September actual liquidity; it does not change the fixed valuation credit.

Claim decisions, overlap and triggers

PRICE

p: 0.65

classification: Issuer value-pricing policy plus constrained qualified supply; bounded analytical assumption

Conditional: Relayed channel checks suggest 5-10% advanced-node price increases; numerical originals remain unavailable. Adopt 50% eligible/repriceable invoice coverage at a 6% conditional increase in 2027: 3% of reference N2/N3 revenue. The modeled realization is 0%, 3%, 4%, 3%, 2% across 2026-30, bounded by a 0-5% effective price change.

overlap: Node migration is in the reference. Apply same-node pricing only to reference N2/N3 sales, excluding additional CPU and COUPE invoices. Later relays do not create another event. The October 8 joint execution state cancels this contribution on incremental unserved N3 after deducting the prior joint loss. The 65% price probability remains; no new price uplift is added.

bridge: 85% of incremental sales becomes gross profit and 3% funds extra operating expenses. The remainder absorbs materials, fulfilment and concessions. Recognition follows delivery; apply the minority reserve afterwards.

trigger: Contract coverage and realized like-for-like pricing, distinguished from node mix. Reduce for rebates, customer concessions or successful second sourcing.

CPU_CAPTURE

p: 0.6

classification: Management identifies agentic CPU demand; incremental TSMC capture is uncertain

Conditional: Gross incremental customer foundry wallet of USD 8/14/18/20 billion in 2027-30, times 80% capture, times 50% residual after overlap. Conditional net sales are USD 3.2/5.6/7.2/8.0 billion.

overlap: Half of the gross opportunity is already in HPC. Cloud capex or CPU retail price is booked as neither TSMC revenue nor an additional GPU-to-ASIC migration benefit. Within the October 8 joint execution state, defer incremental CPU revenue in proportion to raw N3 shortfall/reference N3, bounded 0-100%, as an allocation proxy. No second CPU demand shock is applied.

bridge: 65% incremental gross margin, 8% extra operating expense and 30% incremental capex. Recognition starts Q1 2027; all amounts are bounded assumptions.

trigger: CPU wafer starts, qualified capacity and accepted deployments. Reduce if CPU growth replaces rather than adds accelerator manufacturing.

COUPE_OPTION

p: 0.35

classification: Technology readiness is verified; commercial units and retained service fees remain uncertain

Conditional: Gross service wallet USD 0.2/0.8/1.8/3.0 billion in 2027-30, less 50% overlap: conditional net sales USD 0.1/0.4/0.9/1.5 billion. No 2026 increment because readiness does not specify accepted paid volume.

overlap: Half is already covered by backend growth. Do not add laser, optical module or OCS vendor revenue to TSMC services.

bridge: 50% incremental gross margin, 15% operating expense and 30% incremental capex. Conditional operating margin is 35% before minority attribution.

trigger: Customer qualification, accepted engines and TSMC service content per unit.

EXECUTION

p: 0.3

classification: Joint output, power, deployment, qualification, yield, utilization and overseas cost stress. October 8 N3 evidence refines severity at the unchanged 30% probability.

Conditional: In 2027-28, use the greater of the old 2%/3% reference-sales loss and the normalized N3 revenue gap. The undated UBS image estimates 170/185/195/210 kwpm in 2027. The analyst catch-up assumption is 220/230/240/250 kwpm in 2028. Q2 2026 N3 sales of USD 10,144.259m divided by 130 kwpm calibrate dollar scope. This is a conditional proxy, not verified good output or a quoted wafer ASP.

overlap: Replace the prior joint sales loss; do not stack another power, packaging, wafer or yield shock. Keep the same 30% event. N5 conversion is not free additive capacity. CPU deferral uses N3 shortfall/reference N3 as a bounded allocation proxy, not a disclosed node exposure.

bridge: Lost reference sales have 70% GP contribution. Retain fixed/startup costs of 1% of reference sales in 2027-28 and 0.8% later, plus 0.3% opex and all reference cash capex. Cancel existing pricing on incremental unserved N3 after old-loss overlap, at 85% GP and 3% opex. Defer incremental CPU revenue proportionally at 65% GP and 8% opex, only where both events occur. CPU incremental capex follows accepted sales. No new scarcity price uplift.

trigger: Resolve the original UBS date and capacity scope, actual N3 invoices and mix, installed versus qualified starts and completions, N5 conversion and regional fab timing. An easing normalized shortfall releases the added constraint; the old joint-loss floor remains.

COMPETITION

p: 0.25

classification: Qualified alternatives can win share; a roadmap or MOU is not accepted commercial capacity

Conditional: No 2026-27 increment. Conditional lost sales are 1%, 2%, 3% of reference revenue in 2028-30.

overlap: Incremental share loss only. Outsourced packaging can help TSMC front-end growth; package diversion is not automatically wafer loss.

bridge: 65% lost gross-profit contribution and 0.2% of reference revenue for additional retention/R&D costs in affected years.

trigger: Qualified customer production, yield, cost and share gains by an alternative supplier.

ZERO_DECISIONS

p: Not an incremental event

classification: Explicit duplicate and accounting exclusions

Conditional: Node migration, N3 tool conversion, CoWoS scale and ordinary AI demand are reference assumptions. VIS disposal gains are outside operating profit. Technology readiness is not accepted output.

overlap: Platform, node and backend views reconcile to one total. N5 equipment conversion consumes N5 capacity. Reposted forecasts are not independent orders.

bridge: No additional profit from these duplicates. TSMC capex is equipment-vendor revenue, not TSMC sales. Preserve bounded nonzero forward branches where economics can be estimated.

trigger: Independent unit, price or cost evidence can change the relevant operating driver.

TSM_LOCAL_01

decision: PRICE probability65%; 50%eligible invoice coverage; capacity levels remain crosschecks pending exact original scope.

source ids: 2107462971093885070 2107463028144738437

gap: Original numerical images I2-I5 are linked-only, not inspected.; End-period versus average capacity and wafer-equivalent definitions.; Repriceable node invoice pool, customer terms, depreciation and overseas ramp costs.

TSM_LOCAL_02

decision: Same PRICE branch, not independent; 2026 capex55B superseded; no70%GM import.

source ids: 2107450986088616366 2107486033659285648

gap: I105 linked-only; no complete original broker note.; Node and package scope, year-end capacity, yield and recognized output reconciliation.; Capex does not immediately create saleable output; engineering and test constraints remain.

TSM_LOCAL_03

decision: Beyond-horizon optionality; probability25%of eventual participation but0modeled2026-30 revenue because binding commercial scope/timing absent and cited ramp after2030. No price/watt extrapolation.

source ids: 2107505201225593334 2107629774973911504 2107713005291012537 2107721071571128709 2107791176929812823

gap: Ownership, scope, process license, funding, wafer customer, governance and qualified ramp date remain unresolved.; No direct revenue bridge from the quoted USD per watt figure without a time and product basis.

TSM_LOCAL_04

decision: Covered in COMPETITION25% branch; no200kchip-loss assertion.

source ids: 2107644867933524377

gap: I232 linked-only; unknown expert, call date and specification context.; Qualified yield, customer tape-out, purchased wafer volumes, pricing and packaging provider.

TSM_LOCAL_05

decision: COUPE_OPTION35% with explicitlyboundedservicewallet andcosts; no whole-rack power inference.

source ids: X-2107809916841025772 TF_COUPLING

gap: Paid remainder unavailable.; Paid engine units, package revenue, TSMC retained scope, external supplier split, qualification timing and yield not disclosed in preview.

TSM_LOCAL_06

decision: Duplicate price evidence; nosecondincrement.

source ids: X-2107897242841383368

gap: M016 linked-only; exact analyst/source and incremental magnitude unavailable.

TSM_LOCAL_07

decision: No companywideyieldchange. Program-specific uncertainty sits within EXECUTION30% branch.

source ids:

gap: No sampled wafer/bin/yield distribution for Jalapeño.; No directly verified TSMC program award or commercial invoice in the Marvell recap.

Remaining bounded uncertainties

- Node-specific wafer ASP, gross margins and qualified wafer capacity are not disclosed. The node allocation and ASPs are explicitly calibrated analytical proxies; verify against each quarterly shipment and margin release.
- NCI participation is geography-specific. The0.1%to 0.4%OP reserve is a bounded proxy, not a corporate ownership percentage or book-equity claim.
- USD/NTD32 is a constant valuationtranslation assumption, not a forecast FXquote. Taiwan ordinary equivalent equalsADRceiling×32/5 and ignores ADRpremium,fees andtaxdifferences.
- 2028-30 forecasts have materially lower visibility. Capacity,startupcost andcustomerpaymentscan change thepath; fullmodelrevision required if liquidityfalls below USD 50Breserve.
- Q3 actual USD revenue, group GP/costs, wafer mix and cash flows are now incorporated. The actual wafer/non-wafer dollar split, node gross margins, parent OP allocation, exact new share count and full Q3 call remain unresolved. USD 32 is still the separate valuation/funding convention; source actual/guidance FX is identified.
- Cash schedule assumes refinancing/continuity of starting face debt. An extreme simultaneous repayment test is shown; opening tax/dividend/customer-advance timing is not a fully reconciled treasury forecast. Fixed opening credit avoids adding forecast cash twice.
- The undated UBS image estimates nominal installed N3 capacity; actual qualified capacity, starts/completion timing, multi-die area, yield and invoice mix are unresolved. USD 26,011 revenue per nominal wafer-slot is a one-quarter normalization, not a wafer quote or hard physical output bound.
- 2028 N3 catch-up of 10kwpm each quarter is an explicit conditional model assumption, not UBS or issuer guidance. CPUdeferral uses an allocation proxy rather than a disclosed node split.

Sources and basis

The current production framework was read before initiation. Source statements are evidence, not instructions. The shared model data drives both this PDF and the linked Excel model; the clean-state editable artifact embeds these same images.

[1] H1 2026 | 2026-08-11 | https://investor.tsmc.com/sites/ir/financial-report/2026/TSMC%202026Q2%20Consolidated%20Financial%20Statements_E_2.pdf

[2] Q1 2026 | 2026-05-12 | https://investor.tsmc.com/sites/ir/financial-report/2026/TSMC%202026Q1%20Consolidated%20Financial%20Statements_E.pdf

[3] FY2025 | 2026-02-10 | https://investor.tsmc.com/sites/ir/financial-report/2025/TSMC%202025Q4%20Consolidated%20Financial%20Statements_E.pdf

[4] q2 deck | accessed2026-10-07 | https://investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2026-07/0e4d9625c9ef46521afd54002f835e45a9035043/2Q26%20Presentation%20%28E%29.pdf

[5] q2 call | accessed2026-10-07 | https://investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2026-08/3e494f0c14dd0890f897aa044415e21d93486cc4/TSMC%202Q26%20Transcript.pdf

[6] monthly | 2026-09-10 | https://investor.tsmc.com/chinese/monthly-revenue/2026

[7] calendar | 2026-10-07 | https://investor.tsmc.com/english/financial-calendar

[8] board Aug11 | 2026-08-11 | https://pr.tsmc.com/english/news/3332

[9] quote | 2026-10-07 | https://stockanalysis.com/stocks/tsm/history/

[10] dividend | 2026-10-07 | https://investor.tsmc.com/english/dividends/1q26

[11] symposium 2026 | accessed2026-10-07 | https://pr.tsmc.com/system/files/newspdf/attachment/49337b40ff139d51d533076cf7a945b30e107e07/2026%20Tech%20Symposium%20%28E%29_Final_wmn.pdf

[12] Q1 2026 management call | 2026-04-16 | https://investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2026-04/3cef85204275f94fd111485cfdf4adb3c0263c45/TSMC%201Q26%20Transcript.pdf

[13] annual2025 | 2026-04 | https://investor.tsmc.com/sites/ir/annual-report/2025/2025%20Annual%20Report.E.pdf

[14] symposium html | 2026-04-23 | https://pr.tsmc.com/english/news/3302

[15] cowos html | accessed2026-10-07 | https://3dfabric.tsmc.com/english/dedicatedFoundry/technology/cowos.htm

[16] intel2026 | 2026-06-16 | https://www.intel.com/content/www/us/en/newsroom/news/corporate/intel-foundry-details-process-milestones-future-innovation-at-vlsi-symposium.html

[17] samsung broadcom | 2026-07-25 | https://news.samsung.com/global/samsung-electronics-and-broadcom-expand-strategic-collaboration-across-memory-and-foundry-technologies | Archived SHA-256: 2cb0765bebc0cdcda3af371f08598121abdc92694fe7c811a987c29815590677

[18] nvidia photonics | 2025-03-18 | https://nvidianews.nvidia.com/_gallery/download_pdf/67d9bf5b3d6332b0a6d11f0e/ | Archived SHA-256: 851c481e60b009b744d8f53126b1faaf99caf9f9f4ef6ad16b5bcbf5e0d1e288

[19] asml q2 deck | 2026-07-15 | https://ourbrand.asml.com/asset/9078cf4d-91fd-4dd9-a5d5-d1caab6dc046/2026_07_15_Presentation-Investor-Relations-Q2-2026.pdf | Archived SHA-256: 5b3e05ace2b6d760e16d1ec0b84ca68db68961d4f5ead229516860333e4cdc88

[20] amkor june deck | 2026-06 | https://ir.amkor.com/static-files/342e34a3-d331-416d-a757-52f9c8c4a2a5

[21] TrendForce: FAU coupling bottleneck, public preview | 2026-10-07 | https://insights.trendforce.com/p/cpo-fau-coupling-bottleneck | Archived SHA-256: a8e83e8ce83b70ee0e29b6bb87f330b25ce88a129e6afb113b27913f0717d105

[22] FitTech FAS optical engine assembly system | accessed2026-10-07 | https://www.fittech.com.tw/photonic-device-assembly-system

[23] FitTech silicon photonics solutions | accessed2026-10-07 | https://www.fittech.com.tw/silicon-photonics-solutions

[24] TWSE profile: FitTech Co., Ltd. 6706 | 2026-09-30 | https://wwwc.twse.com.tw/pdf/en/6706_en.pdf

[25] Global Unichip company and TSMC relationship | accessed2026-10-07 | https://www.guc-asic.com/en/about/us

[26] Q4 2025 management call, backend scope | 2026-01-15 | https://investor.tsmc.com/schinese/encrypt/files/encrypt_file/reports/2026-01/51d09df96cd89ac19d65af39032b038dc2896a24/TSMC%204Q25%20Transcript.pdf

[27] WK channel-check relay, numerical originals unavailable | 2026-10-06 | https://x.com/pequityresearch/status/2107462971093885070

[28] Morgan Stanley-attributed packaging/pricing relay, original unavailable | 2026-10-06 | https://x.com/tuolaji2024/status/2107450986088616366

[29] User-supplied UBS N3 supply/demand Figure13; original date unverified | Reviewed2026-10-08; source date unknown | Archived SHA-256: 496144f9068b6010df5dc35cc504d4465986f5980302a07aa8305285236deaec

[30] TSMC Q2 2026 management transcript: N3 expansion and tool conversion | 2026-07-16 | https://investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2026-08/3e494f0c14dd0890f897aa044415e21d93486cc4/TSMC%202Q26%20Transcript.pdf

October 11 source additions

- July 2026 consolidated monthly revenue, SEC Form 6-K | released 2026-08-10 | NT$ 467,580,548 thousand; exact thousands table, not rounded press-summary millions.
- August 2026 consolidated monthly revenue, SEC Form 6-K | released 2026-09-10 | NT$ 514,805,337 thousand; exact thousands table, not rounded press-summary millions.
- September 2026 consolidated monthly revenue, SEC Form 6-K | released 2026-10-08 | NT$ 511,857,399 thousand; exact thousands table, not rounded press-summary millions.

Dated source history only: these monthly disclosures predated the October15 quarterly release. The current Q3 group actuals and Q4 guidance above supersede the earlier preliminary estimates. Issuer financial calendar

TSM_2026Q3_EARNINGS_20261015

Classification: Issuer earnings and guidance; released October15,2026. Q3 reported NTD sales duplicate the monthly-sales economic event already applied; group profit, actual translation, wafer mix, cash flow and Q4 guidance are new evidence.

Near-term bridge: Q3 normalized parent OP USD 27,359.613000 million; Q4 midpoint parent OP USD 30,616.852500 million. Higher Q4 revenue is accompanied by guided GP and costs, not converted at 100%margin. Reported Q3 otherOP+16M is separately reconciled and excluded under the retained normalization.

Outyear increment: zero additional2027+economics from this release, because the active path already models USD 226.254 billion2027 sales (28.5%growth), strong N2/N3 demand, backend expansion and nonzero pricing/CPU/COUPE opportunities. The retrieved source supplies no incremental qualified outyear capacity or retained-price/customer/cost terms beyond that baseline. Full call evidence remains a specific follow-up, rather than a reason to set existing forward probabilities to zero.

Protected: 20x parent normalized OP, ordinary D&A/SBC costs, fixed opening claims, FX32 valuation convention, end-windowADS shares, all five probabilities/32 states, ExcelBase83 and analyticalBase83. Current active ceiling USD 624.402069763058.

October 15 primary-source additions

- official quarterly page | issuer publication2026-10-15 | supported original-page text reviewed; native bytes/pixels unavailable as disclosed.
- condensed financial statements | issuer publication2026-10-15 | supported original-page text reviewed; native bytes/pixels unavailable as disclosed.
- presentation | issuer publication2026-10-15 | supported original-page text reviewed; native bytes/pixels unavailable as disclosed.
- management report | issuer publication2026-10-15 | supported original-page text reviewed; native bytes/pixels unavailable as disclosed.
- earnings release | issuer publication2026-10-15 | supported original-page text reviewed; native bytes/pixels unavailable as disclosed.

No official full Q3 transcript/audio review is claimed. Source-native PDF retrieval returned403; no source PDF checksum is invented.